TAIWAN TECH

Computer Vision and Applications

Lecture-03 Projective 2D geometry

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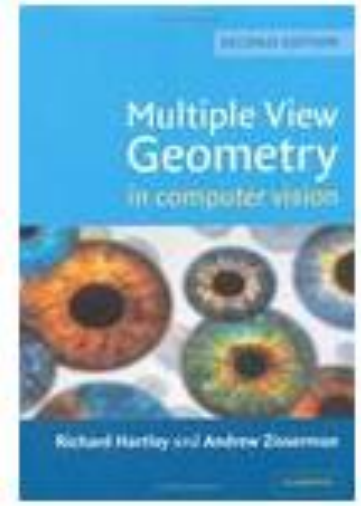
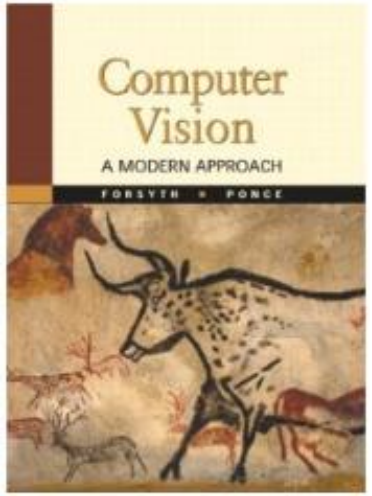
Projective 2D geometry

Lecture Reference at:

- Multiple View Geometry in Computer Vision, Chapter 2. (major)
- Computer Vision A Modern Approach, Chapter 10.





Keywords list

- Scalar, vector, matrix
- Line equation, conics
- Homogeneous, inhomogenous, homogeneous coordinates
- Homogenous represnetation for point, line.
- Projective invariants
- DOF: degree of freedom
- Homography (homographic transform)
- Point at infinity
- Dual, duality, envelopes

Notation remark

- **Bold**: vector or matrix form, ex. $a\mathbf{x} + b\mathbf{y} = \mathbf{c}$
- *Italic*: scalar or variable, ex. $ax+by = 1$
- Upper case for 3D, ex. X
- Lower case for 2D, ex. x
- **X**→ a 3D vector, **x**→ a 2D vector
- $[X, Y, Z, 1]$ → a homogenous 3D vector (consisting of 3 scalars)

- Normal text is usually a list of text or character.

Projective 2D geometry

Topics

- Points, lines & conics
- Transformations & invariants (between images)
- 1D projective geometry and the Cross-ratio

Homogeneous coordinates

- Homogeneous representation of lines

$$ax + by + c = 0 \qquad (a, b, c)^{\mathrm{T}}$$

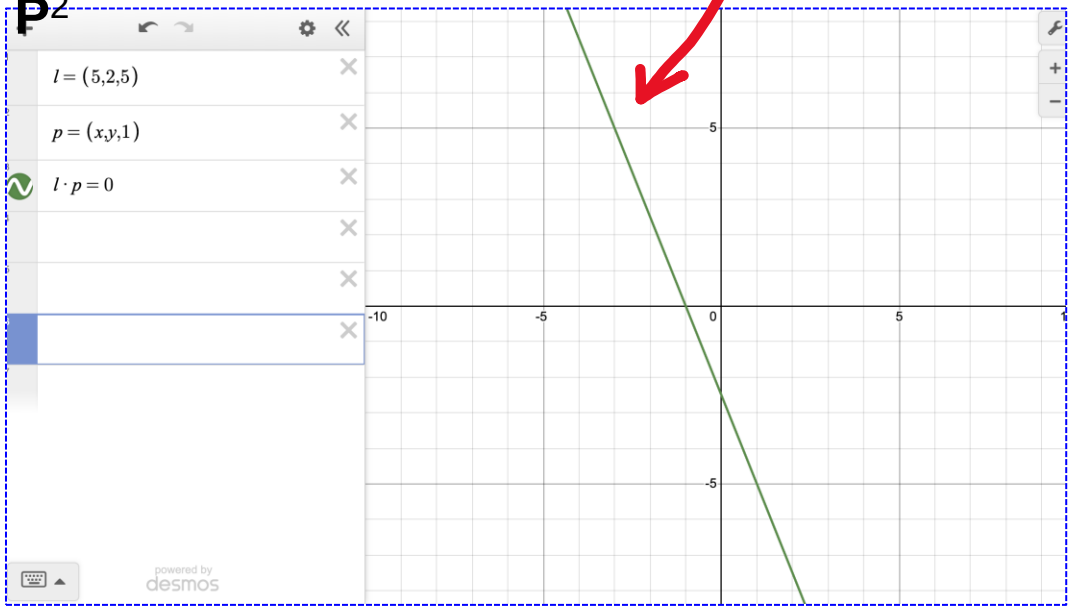
$$(ka)x + (kb)y + kc = 0, if\ k \neq 0 \qquad (a, b, c)^{\mathrm{T}} \sim k(a, b, c)^{\mathrm{T}}$$

$\mathbf{l} = (5, 2, 5)^{\mathrm{T}}$
or 5*x*+2*y*+5=0

Equivalence class of vectors, any vector is representative

Set of all equivalence classes in $\mathbf{R}^3$–$(0,0,0)^{\mathrm{T}}$ forms $\mathbf{P}^2$

Define one line as a vector format: $\mathbf{l} = (a, b, c)^{\mathrm{T}}$



https://www.desmos.com/

Homogeneous coordinates

- Homogeneous coordinates of points

  $\mathbf{x} = (x, y, 1)^{\mathrm{T}}$ on $\mathbf{l} = (a, b, c)^{\mathrm{T}}$ if and only if $ax + by + c = 0$

  Can rewrite as $(x, y, 1) \cdot (a, b, c)^{\mathrm{T}} = (x, y, 1)\mathbf{l} = 0$

  The answer will be $(x, y, 1)^{\mathrm{T}} \sim k(x, y, 1)^{\mathrm{T}}, \forall k \neq 0$

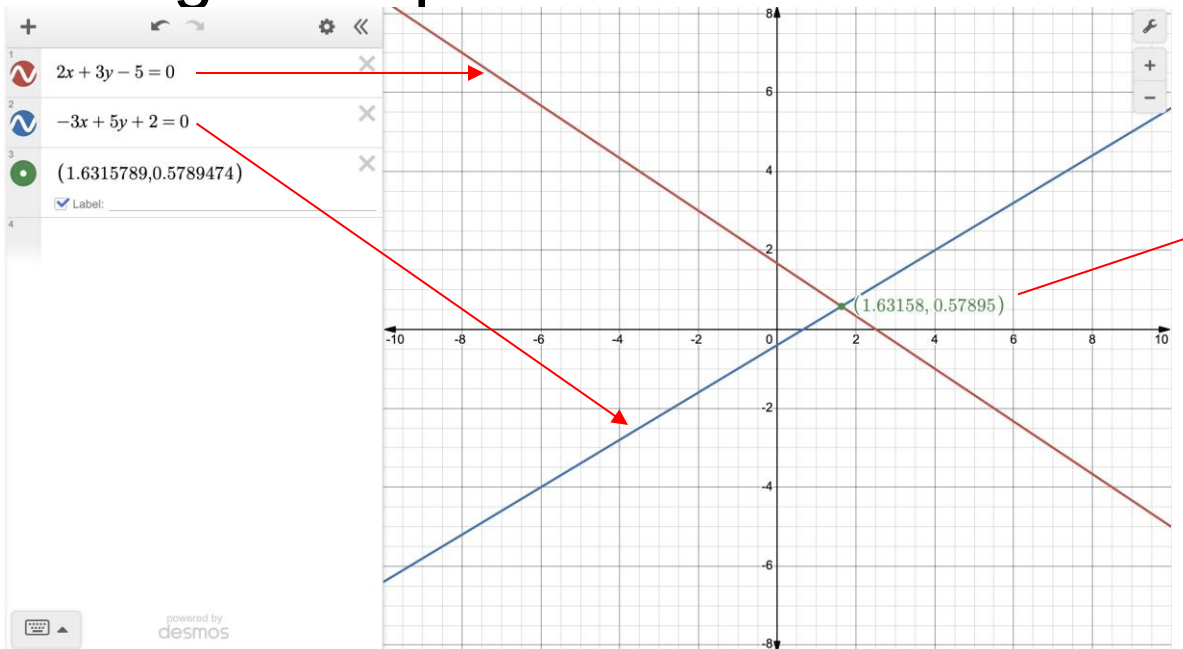
- The point **x** lies on the line **l** if and only if $\mathbf{x}^{\mathrm{T}}\mathbf{l} = \mathbf{l}^{\mathrm{T}}\mathbf{x} = 0$

  Homogenous coordinates $(x_1, x_2, x_3)^{\mathrm{T}}$ but only 2DOF

  Inhomogenous coordinates $(x, y)^{\mathrm{T}}$

2D Points from lines and vice-versa

- Intersections of lines
  - The intersection of two lines **l** and **l'** is $\mathbf{x} = \mathbf{l} \times \mathbf{l}'$
- Line joining two points
  - The line through two points **x** and **x'** is $\mathbf{l} = \mathbf{x} \times \mathbf{x}'$

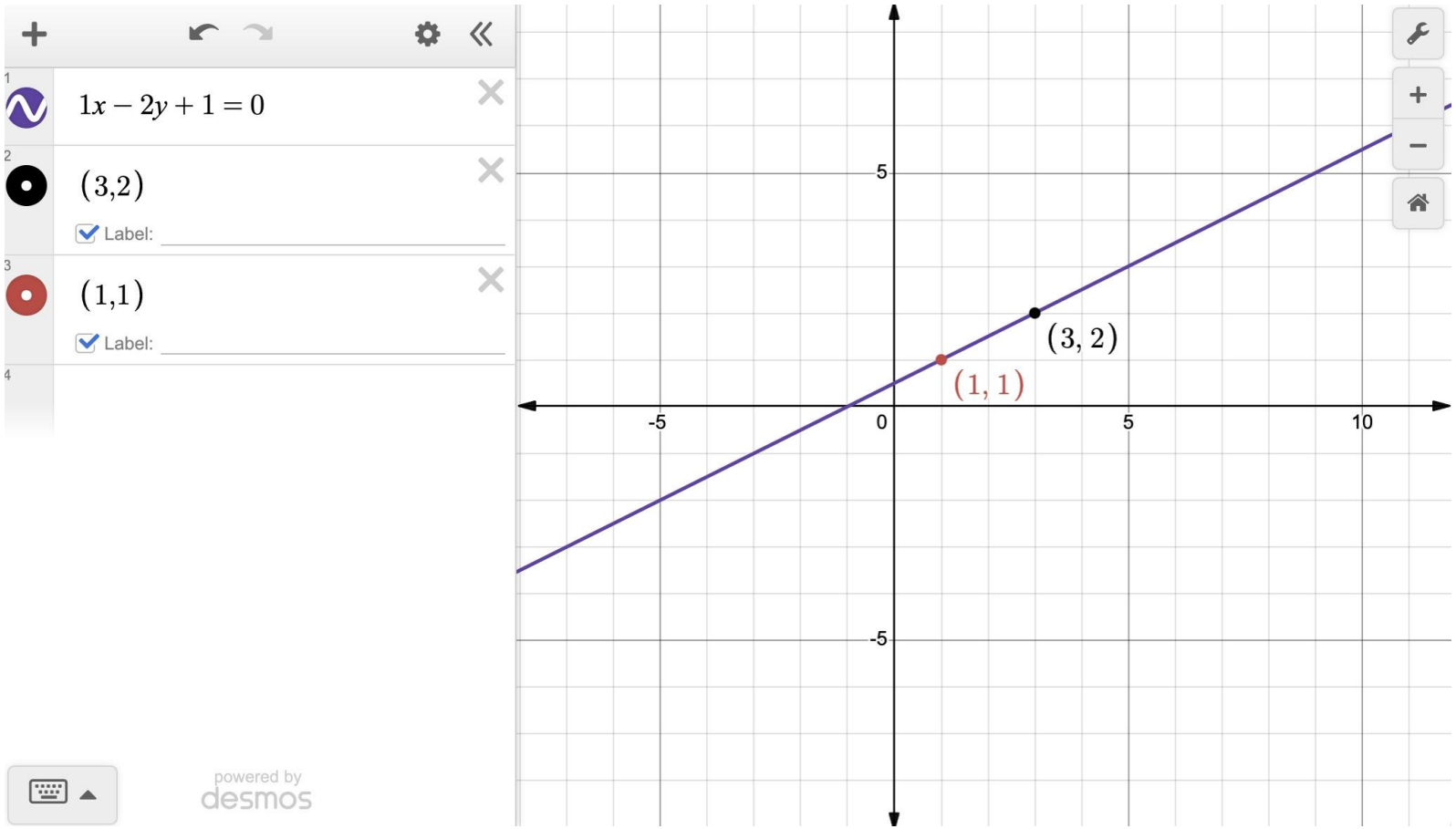


$$\mathbf{x} = \mathbf{l} \times \mathbf{l}'$$

https://www.desmos.com/

2D Points from lines and vice-versa

- Example: Line joining two points



$$\begin{bmatrix}3\\2\\1\end{bmatrix} \times \begin{bmatrix}1\\1\\1\end{bmatrix} = \begin{bmatrix}1\\-2\\1\end{bmatrix}$$

or

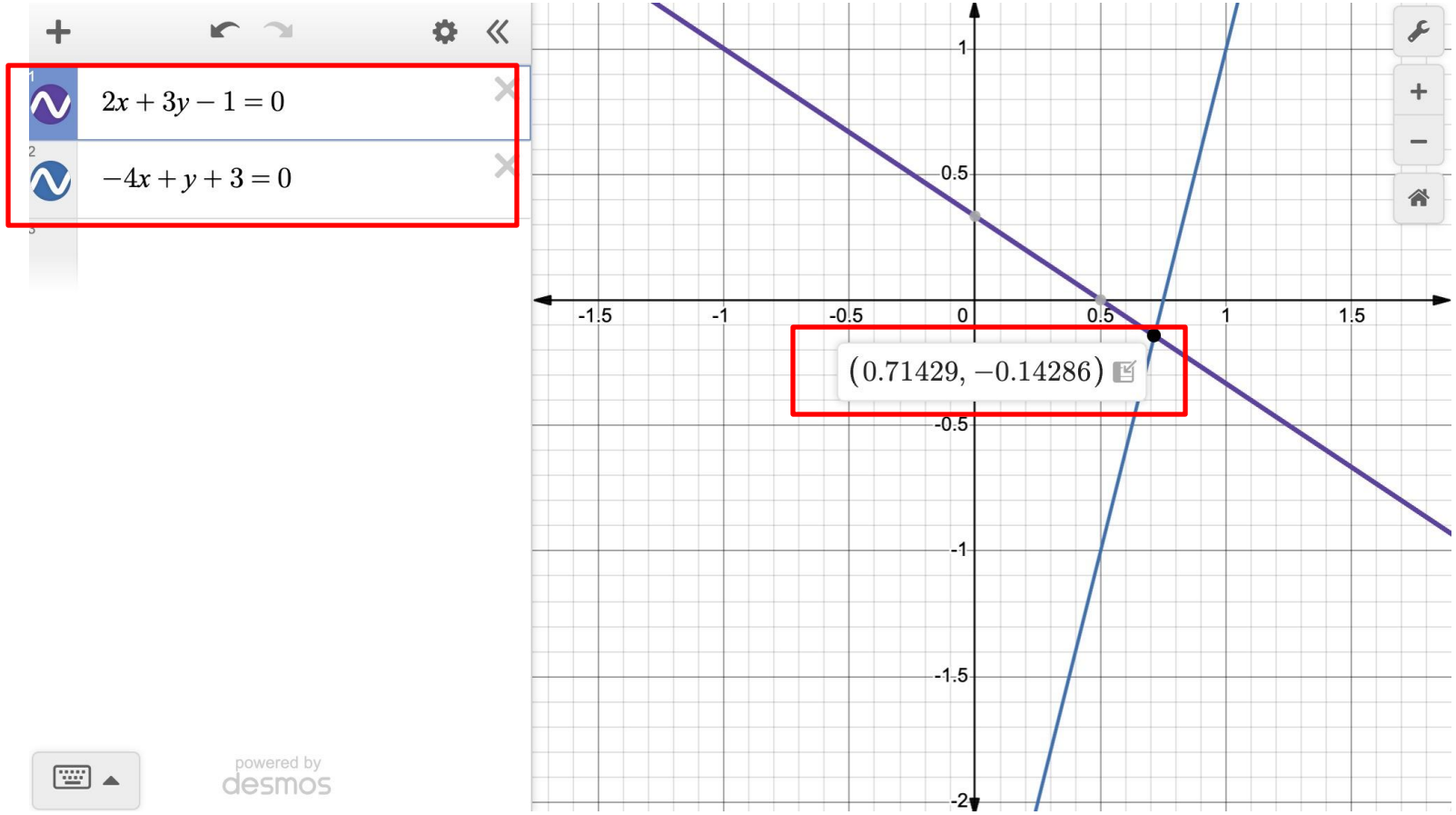
$$\begin{bmatrix}1\\1\\1\end{bmatrix} \times \begin{bmatrix}3\\2\\1\end{bmatrix} = \begin{bmatrix}-1\\2\\-1\end{bmatrix}$$

$$\mathbf{x} \times \mathbf{x}' = \mathbf{l}$$

https://www.desmos.com/

2D Points from lines and vice-versa

- Example: Line intersection example



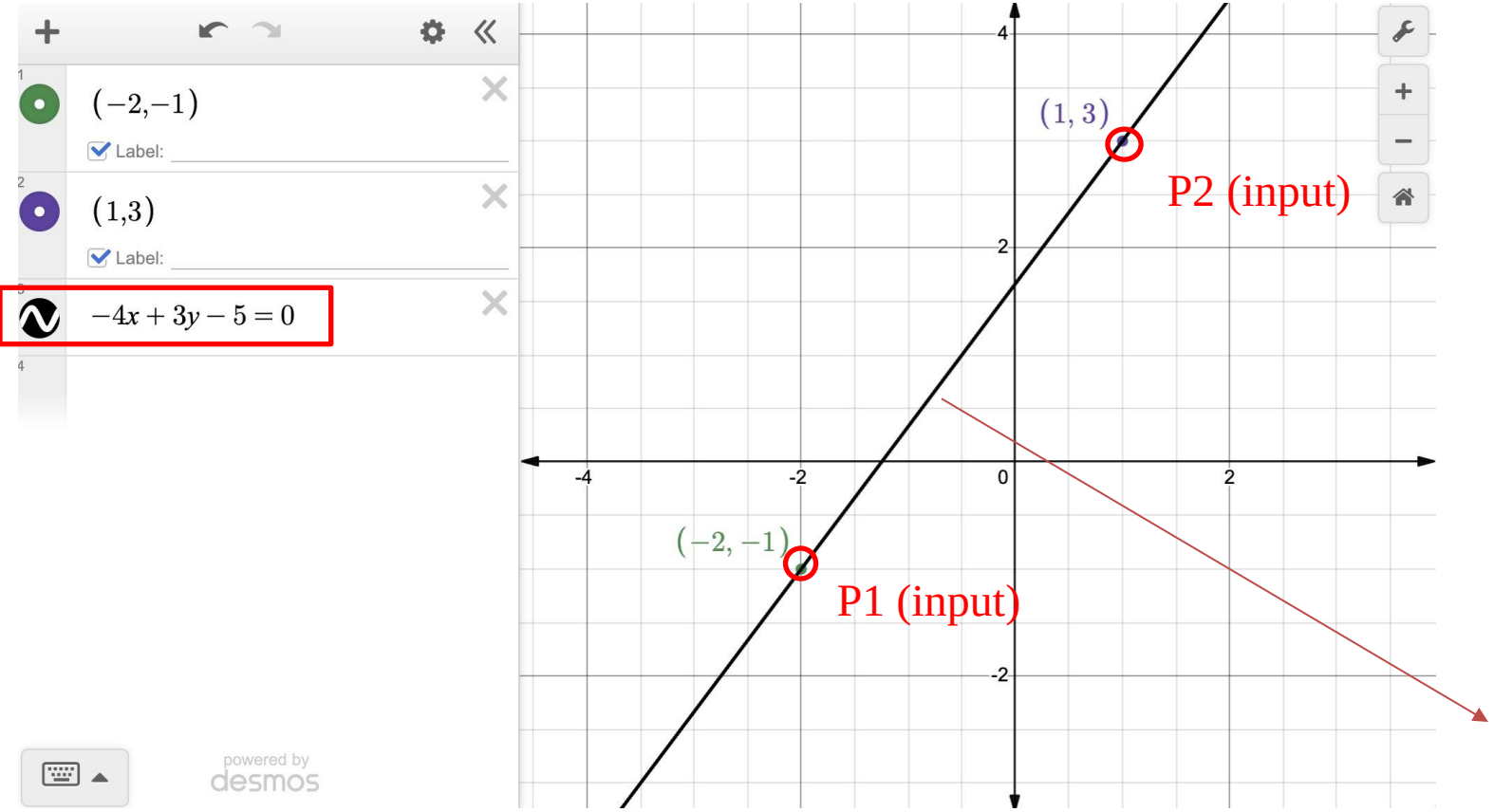
Scilab 6.1.1 Console

```
--> l1=[2 3 -1]'
 l1  =

   2.
   3.
  -1.

--> l2=[-4 1 3]'
 l2  =

  -4.
   1.
   3.

--> p=cross(l1,l2)
 p  =

   10.
  -2.
   14.

--> p./p(3)
 ans  =

   0.7142857
  -0.1428571
   1.
```

https://www.desmos.com/

2D Points from lines and vice-versa

- Exampel: to form a line from two points



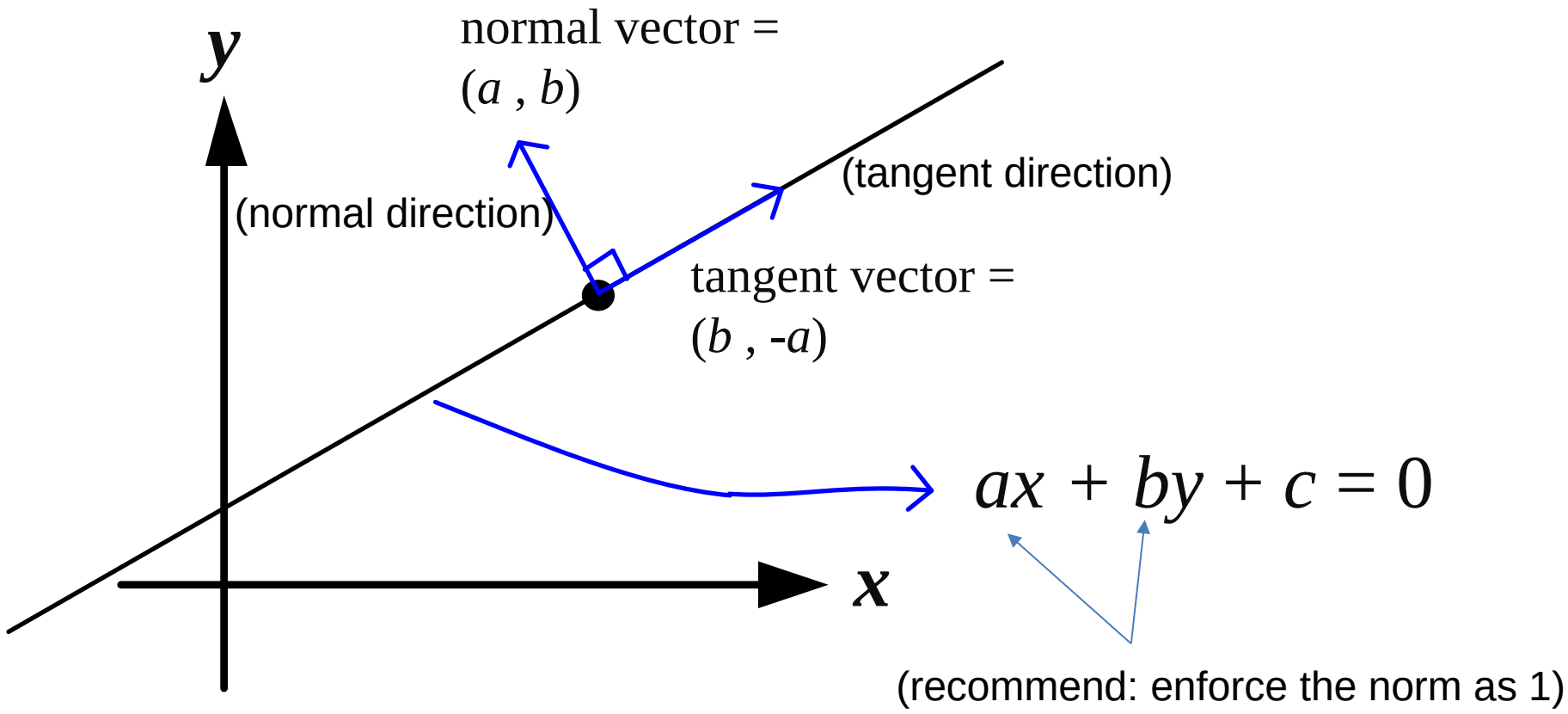

```
Scilab 6.1.1 Console

--> p1=[-2 -1 1]'
 p1  =

  -2.
  -1.
   1.

--> p2=[1 3 1]'
 p2  =

   1.
   3.
   1.

--> l=cross(p1,p2)
 l  =

  -4.
   3.
  -5.
```

Line equation (output)

https://www.desmos.com/

Points from lines and vice-versa

- Normal vector and tangent vector of one line:



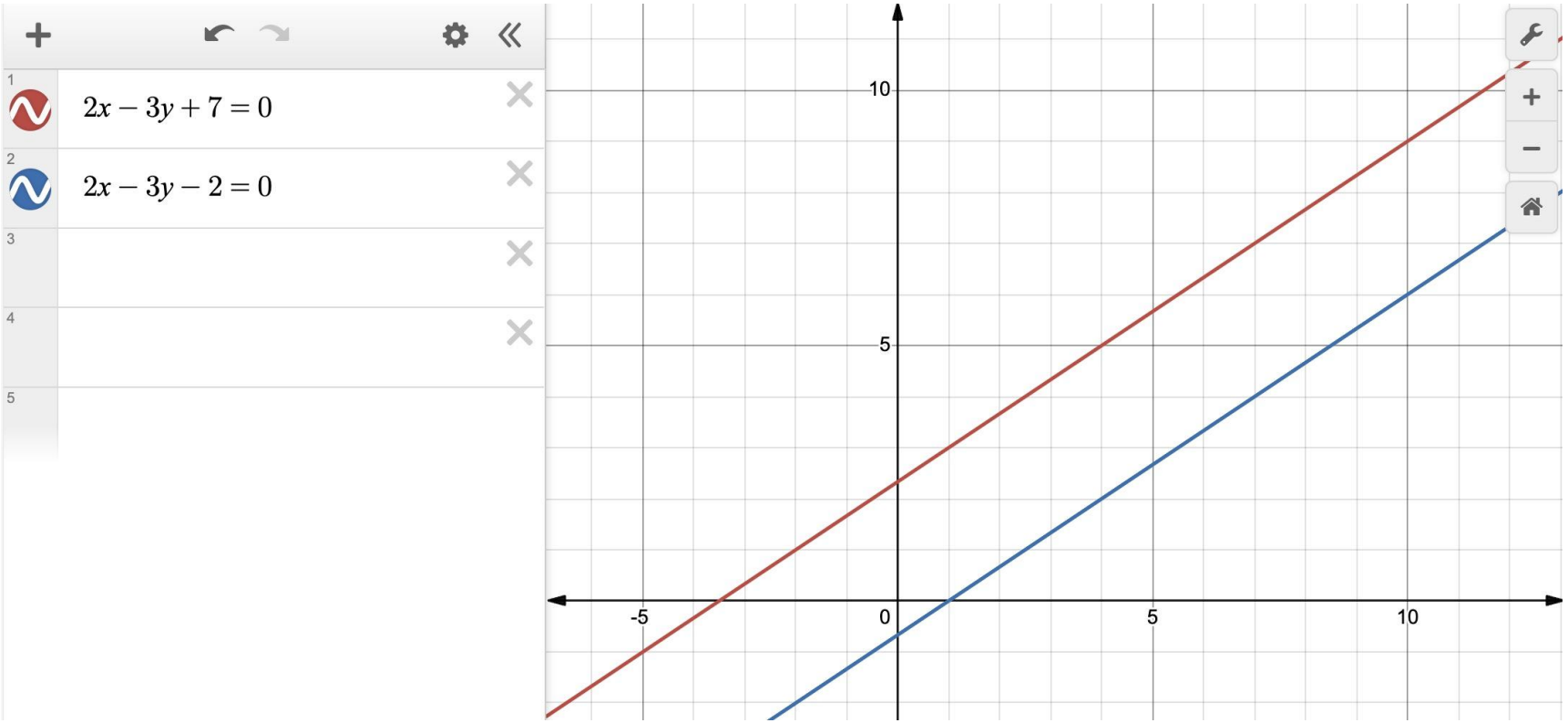
$$ax + by + c = 0$$

(recommend: enforce the norm as 1)

Ideal points and the line at infinity

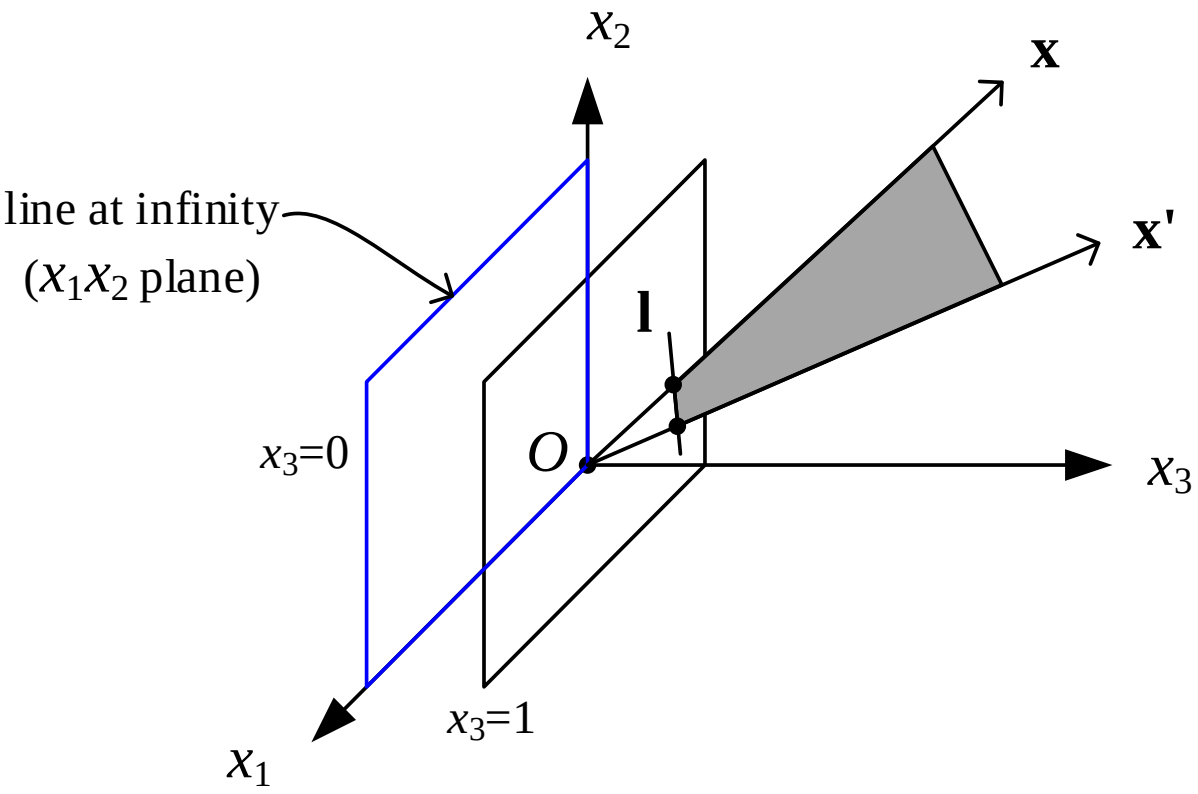
- Intersections of parallel lines

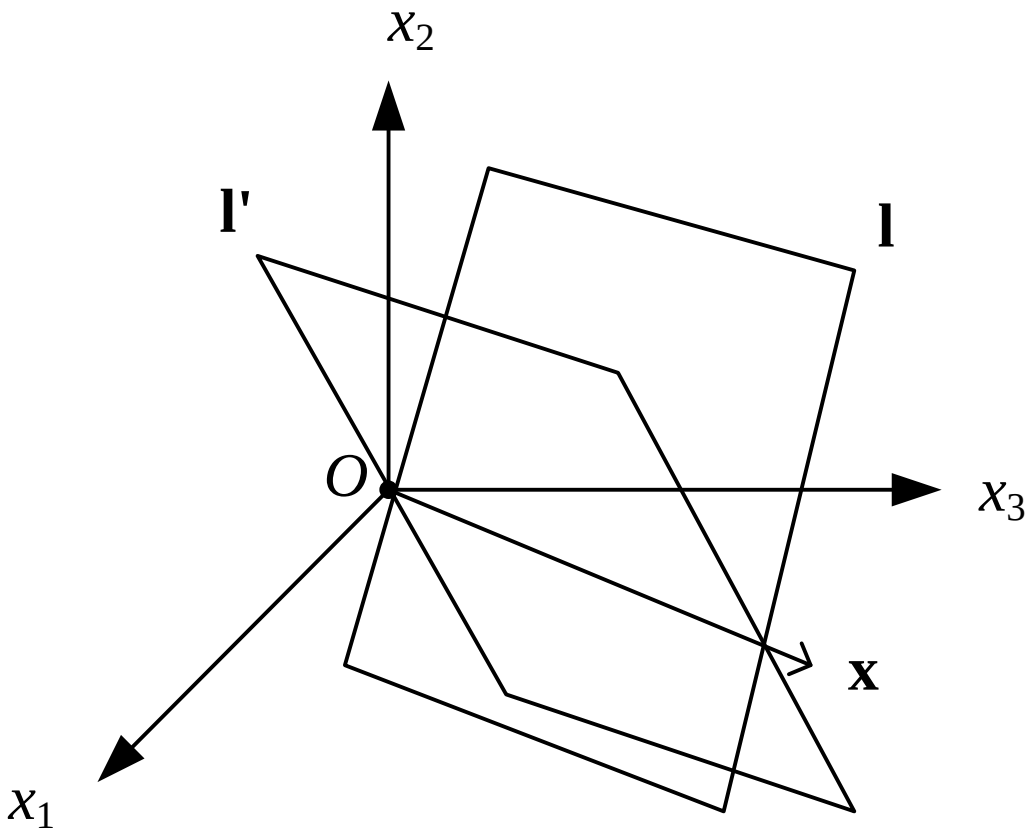
Ex. $\mathbf{l} = (a, b, c)^{\mathrm{T}}$ and $\mathbf{l}' = (a, b, c')^{\mathrm{T}} \rightarrow \mathbf{x} = \mathbf{l} \times \mathbf{l}' = (b, -a, 0)^{\mathrm{T}}$ point at infinity



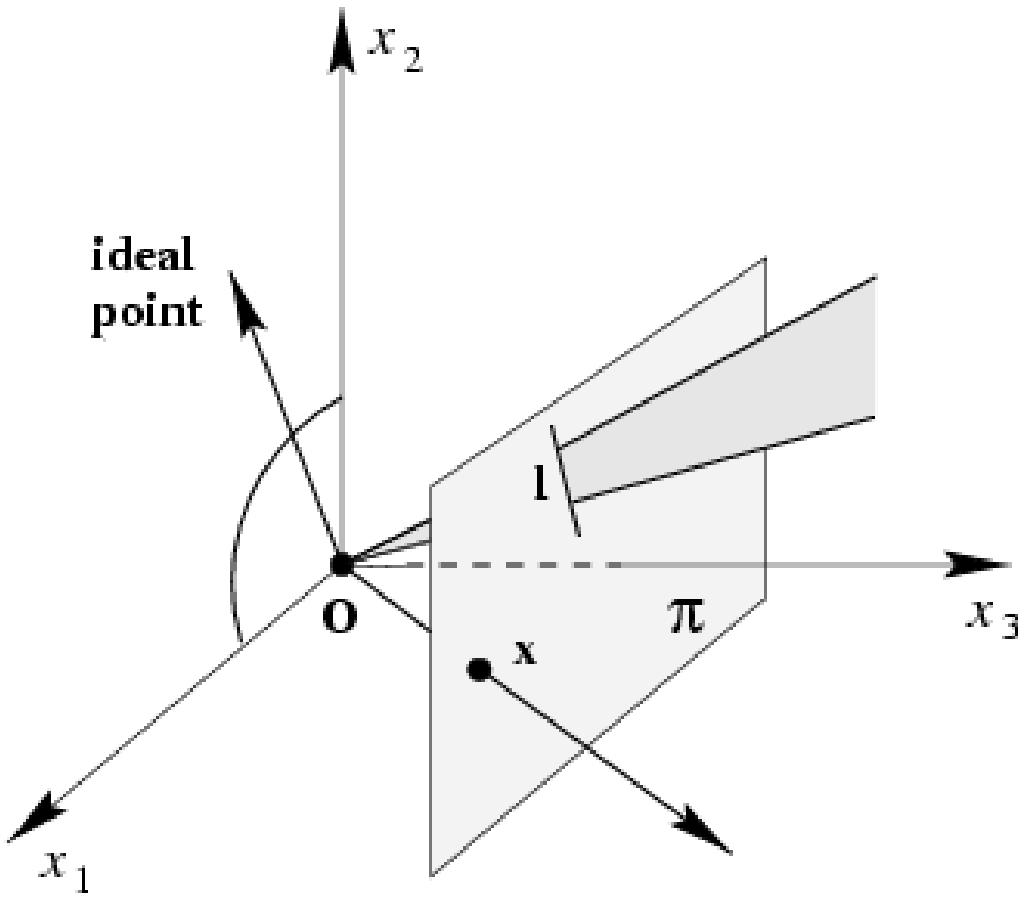
Ideal points and the line at infinity

- Schematic of homogenous coordinates:





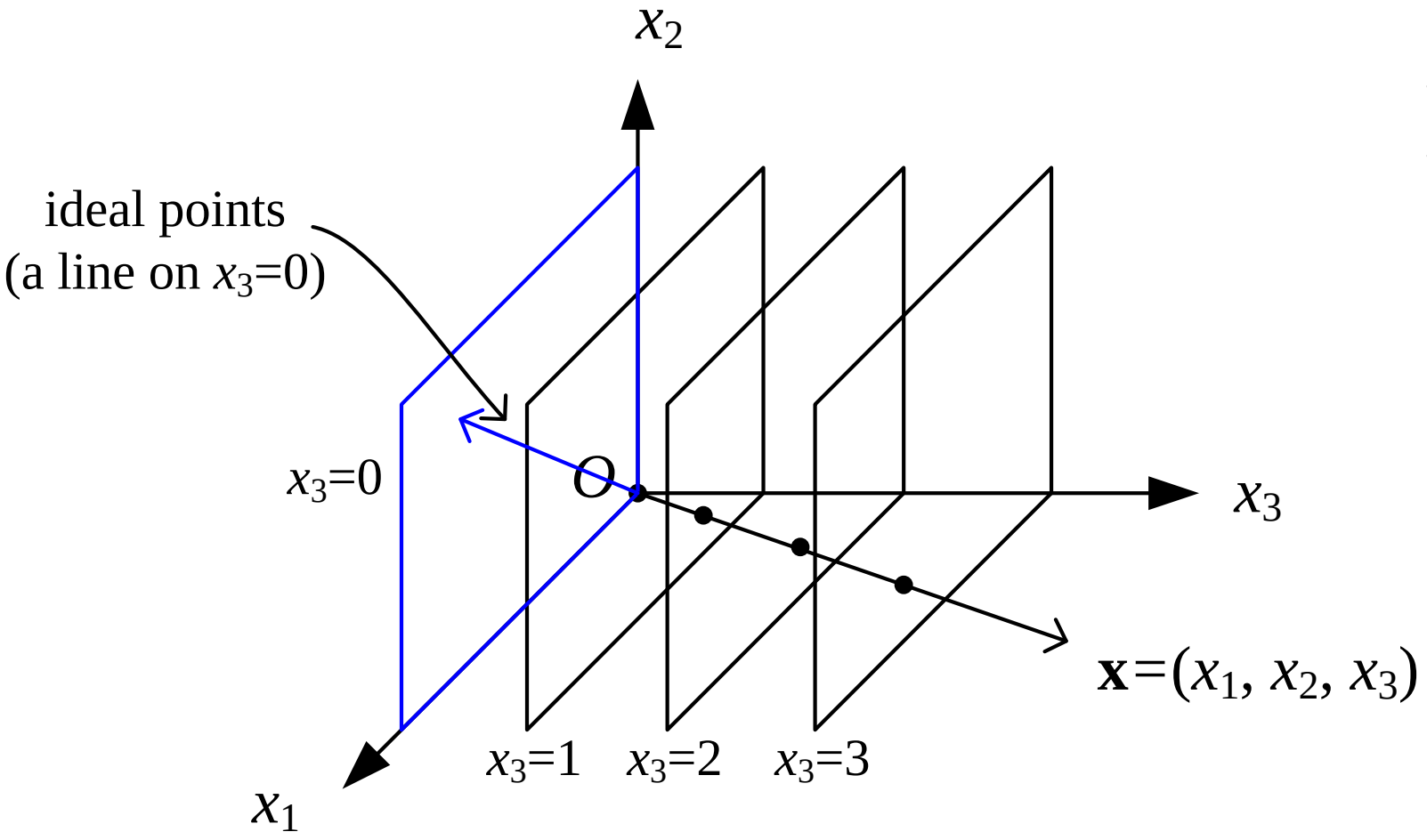
A model for the projective plane



exactly one line through two points

exactly one point at intersection of two lines

A model for the projective plane—cont.



$$\mathbf{x}_\mathrm{a} = (x_1, y_1, 0)^\mathrm{T}$$
$$\mathbf{x}_\mathrm{b} = (x_2, y_2, 0)^\mathrm{T}$$

$$\mathbf{l} = \mathbf{x}_\mathrm{a} \times \mathbf{x}_\mathrm{b}$$
$$= k(0, 0, 1)^\mathrm{T}$$

$$\mathbf{l}_\infty = k(0, 0, 1)^\mathrm{T}$$

Duality of 2D lines and points

Duality principle:

- To any theorem of 2-dimensional projective geometry there corresponds a dual theorem, which may be derived by interchanging the role of points and lines in the original theorem

$$\mathbf{x} \Longleftrightarrow \mathbf{l}$$

$$\mathbf{x}^{\mathrm{T}}\mathbf{l} = \mathbf{0} \Longleftrightarrow \mathbf{l}^{\mathrm{T}}\mathbf{x} = \mathbf{0}$$

$$\mathbf{x} = \mathbf{l} \times \mathbf{l}' \Longleftrightarrow \mathbf{l} = \mathbf{x} \times \mathbf{x}'$$

Conics (polynomial)

- Curve described by 2nd-degree equation in the plane

$$ax^2 + bxy + cy^2 + dx + ey + f = 0$$

or homogenized (replaced according to $x \to x_1/x_3$ and $y \to x_2/x_3$)

$$ax_1^2 + bx_1x_2 + cx_2^2 + dx_1x_3 + ex_2x_3 + fx_3^2 = 0$$

or in matrix form

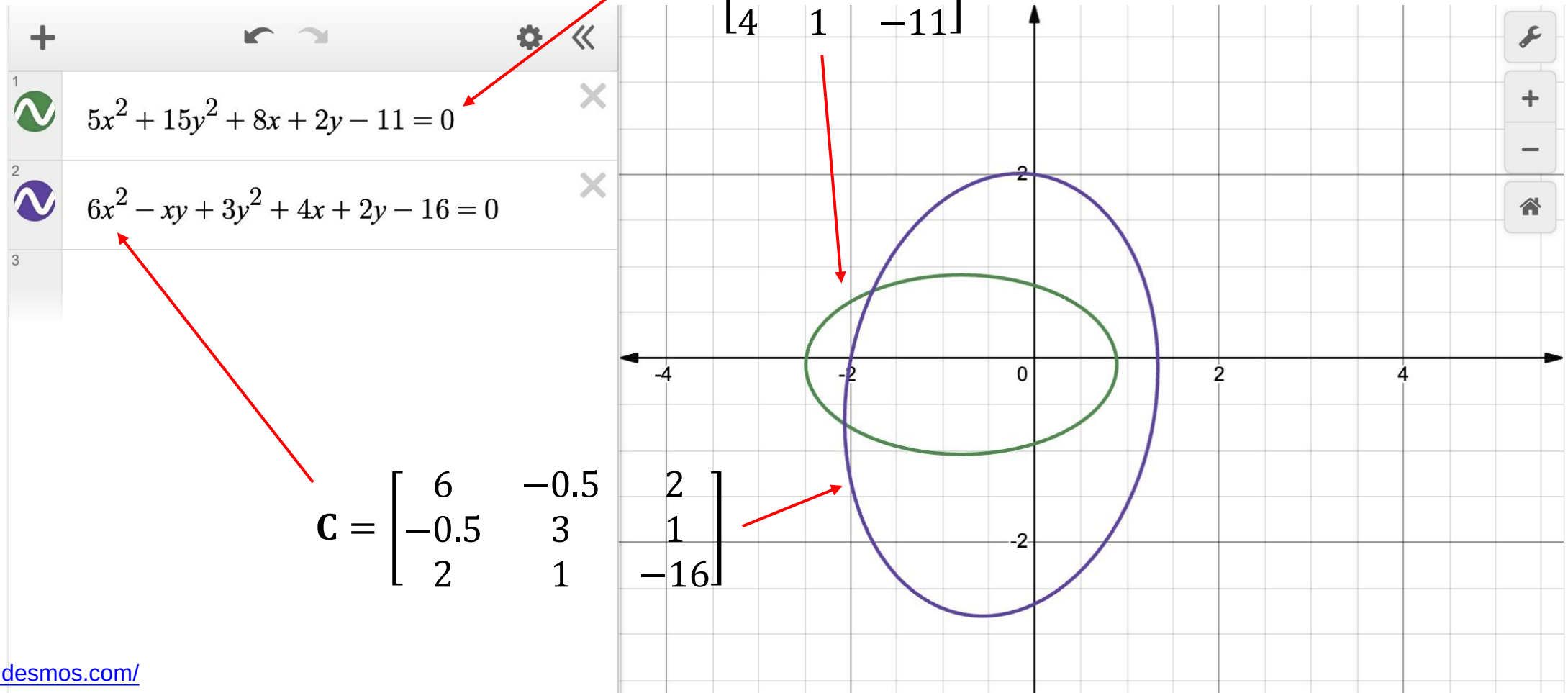
$$[x \quad y \quad 1]\begin{bmatrix} a & b/2 & d/2 \\ b/2 & c & e/2 \\ d/2 & e/2 & f \end{bmatrix}\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = 0 \rightarrow \mathbf{x}^{\mathrm{T}}\mathbf{C}\mathbf{x} = 0$$

where $\mathbf{C} = \begin{bmatrix} a & b/2 & d/2 \\ b/2 & c & e/2 \\ d/2 & e/2 & f \end{bmatrix}$

Conics (polynomial)

- Example

$$\mathbf{C} = \begin{bmatrix} 5 & 0 & 4 \\ 0 & 15 & 1 \\ 4 & 1 & -11 \end{bmatrix}$$



$$\mathbf{C} = \begin{bmatrix} 6 & -0.5 & 2 \\ -0.5 & 3 & 1 \\ 2 & 1 & -16 \end{bmatrix}$$

https://www.desmos.com/

Conics

- Five points define a conic

For each point the conic passes through

$$ax^2 + bxy + cy^2 + dx + ey + f = 0$$

or in matrix form

$$(x^2,\ xy,\ y^2,\ x,\ y,\ 1) \cdot \mathbf{c} = 0, \text{ where } \mathbf{c} = (a, b, c, d, e, f)^{\mathrm{T}}$$

stacking constraints yields

$$\begin{bmatrix} x_1^2 & x_1y_1 & y_1^2 & x_1 & y_1 & 1 \\ x_2^2 & x_2y_2 & y_2^2 & x_2 & y_2 & 1 \\ x_3^2 & x_3y_3 & y_3^2 & x_3 & y_3 & 1 \\ x_4^2 & x_4y_4 & y_4^2 & x_4 & y_4 & 1 \\ x_5^2 & x_5y_5 & y_5^2 & x_5 & y_5 & 1 \end{bmatrix} \cdot \mathbf{c} = 0 \text{ , or } \begin{bmatrix} x_1^2 & x_1y_1 & y_1^2 & x_1 & y_1 \\ x_2^2 & x_2y_2 & y_2^2 & x_2 & y_2 \\ x_3^2 & x_3y_3 & y_3^2 & x_3 & y_3 \\ x_4^2 & x_4y_4 & y_4^2 & x_4 & y_4 \\ x_5^2 & x_5y_5 & y_5^2 & x_5 & y_5 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \\ e \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ -1 \\ -1 \\ -1 \end{bmatrix} where\ f = 1$$

Conics

- Five points define a conic, an example: $0.2x^2 + y^2 - 10 = 0$

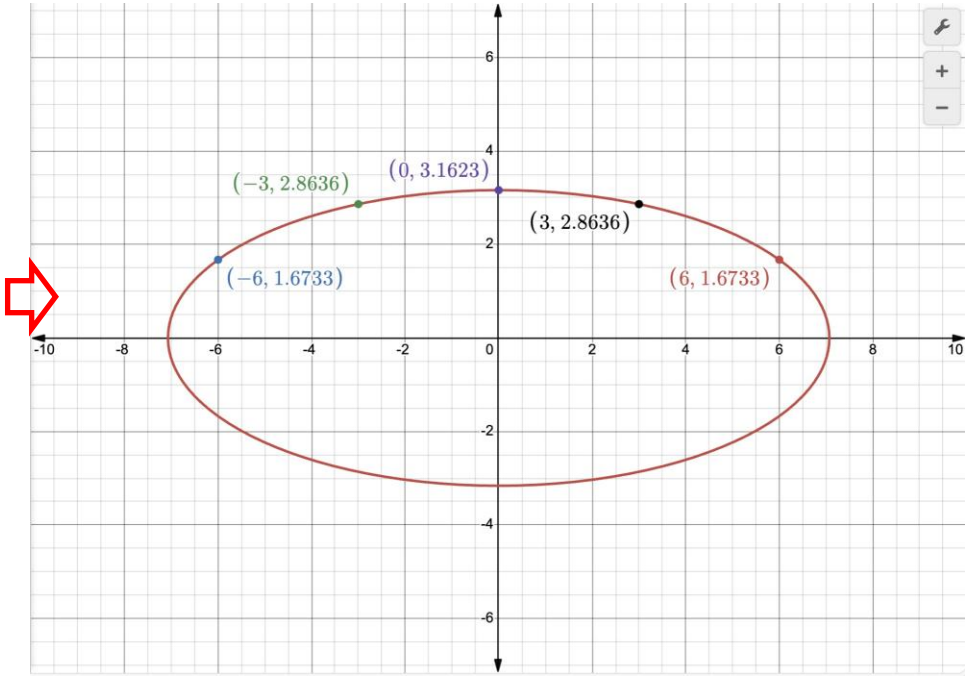
5 points (known as input) determine a conic (unknown, output):

(-6,1.6733,1)
(-3,2.8636,1)
(0,3.1623,1)
(3,2.8636,1)
(6,1.6733,1)

$$\begin{bmatrix} 36.0000 & -10.0399 & 2.8000 & -6.0000 & 1.6733 \\ 9.0000 & -8.5907 & 8.2000 & -3.0000 & 2.8636 \\ 0 & 0 & 10.0000 & 0 & 3.1623 \\ 9.0000 & 8.5907 & 8.2000 & 3.0000 & 2.8636 \\ 36.0000 & 10.0399 & 2.8000 & 6.0000 & 1.6733 \end{bmatrix} * \begin{bmatrix} a \\ b \\ c \\ d \\ e \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ -1 \\ -1 \\ -1 \end{bmatrix}$$

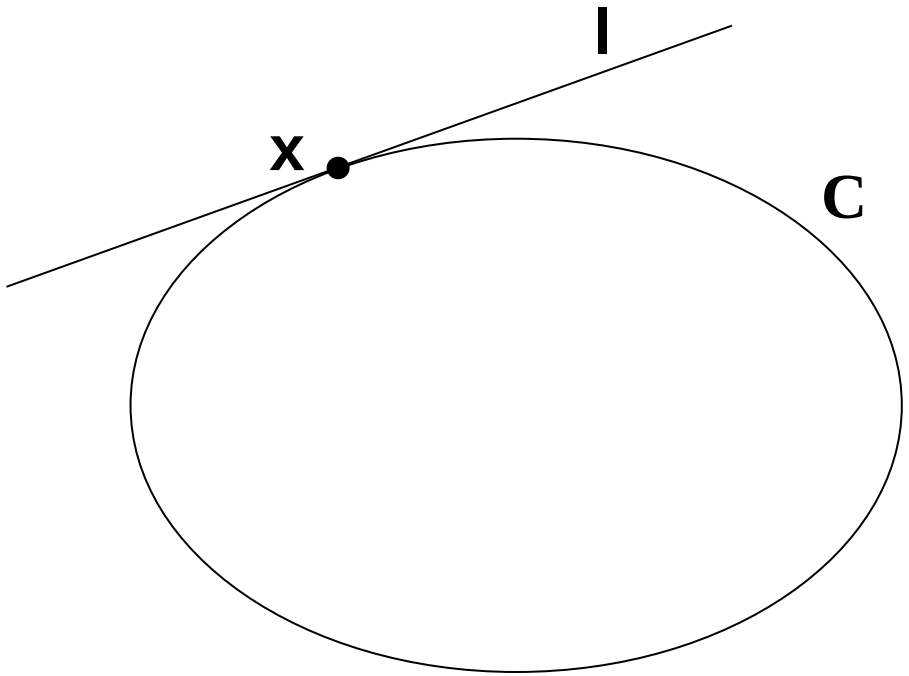
→ Solve it, then get $\begin{bmatrix} a \\ b \\ c \\ d \\ e \end{bmatrix} = \begin{bmatrix} -0.02 \\ 0 \\ -0.1 \\ 0 \\ 0 \end{bmatrix}$

→ $0.2x^2 + y^2 - 10 = 0$



Tangent lines to conics

- The line l tangent to C at point x on C is given by l=Cx



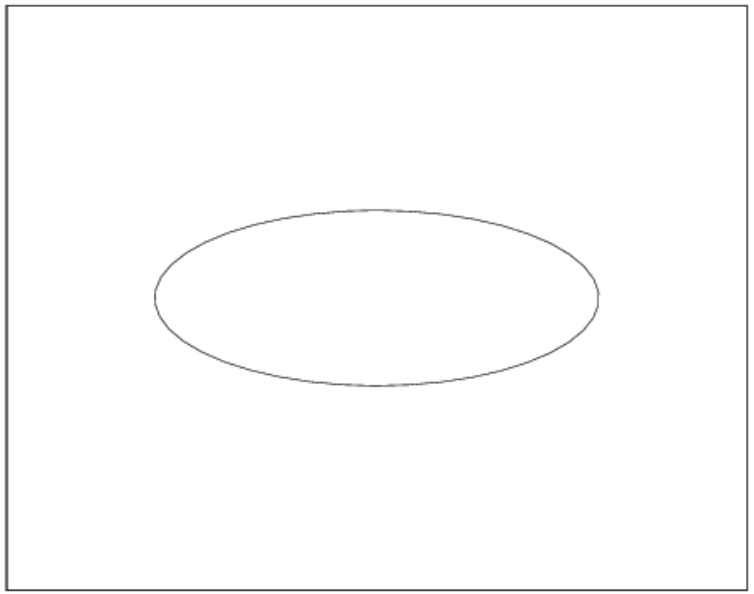
Since $\mathbf{x}^{\mathrm{T}}\mathbf{C}\mathbf{x} = 0 = \mathbf{x}^{\mathrm{T}}\mathbf{l} = \mathbf{l}^{\mathrm{T}}\mathbf{x}$

We get $\mathbf{C}\mathbf{x} = \mathbf{l}$ or $\mathbf{l}^{\mathrm{T}} = \mathbf{x}^{\mathrm{T}}\mathbf{C}$

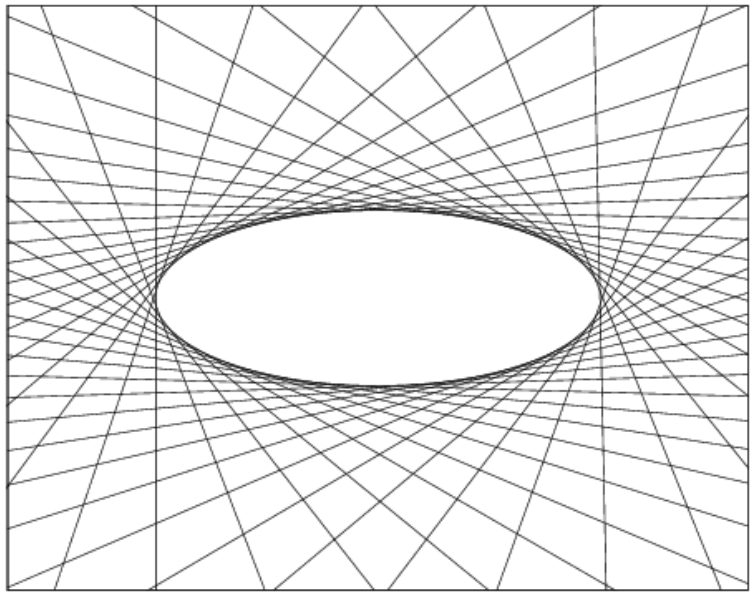
Dual conics

A line tangent to the conic C satisfies $\mathbf{l}^{\mathrm{T}}\mathbf{C}^{*}\mathbf{l} = 0$

- In general ($\mathbf{C}$ full rank): $\mathbf{C}^{*} = \mathbf{C}^{-1}$
- Dual conics = line conics = conic envelopes



$\mathbf{x}^{\mathrm{T}}\mathbf{C}\mathbf{x} = 0$



$\mathbf{l}^{\mathrm{T}}\mathbf{C}^{*}\mathbf{l} = 0$

Dual conics

- A line tangent to the conic **C** satisfies

$$\mathbf{l}^{\mathrm{T}}\mathbf{C}^{*}\mathbf{l} = 0 \rightarrow \mathbf{C}^{*} = \mathbf{C}^{-1}$$

- Proof:

(says tangent points)

Since $\mathbf{x}^{\mathrm{T}}\mathbf{C}\mathbf{x} = 0$, and $\mathbf{l} = \mathbf{C}\mathbf{x}$ (we can get $\mathbf{x} = \mathbf{C}^{-1}\mathbf{l}$)

And line on the conic :

$$\mathbf{x}^{\mathrm{T}}\mathbf{C}\mathbf{x} = \mathbf{0}$$

$$\left(\mathbf{C}^{-1}\mathbf{l}\right)^{\mathrm{T}}\mathbf{C}(\mathbf{C}^{-1}\mathbf{l}) = \mathbf{0} \rightarrow \mathbf{l}^{\mathrm{T}}\mathbf{C}^{-\mathrm{T}}\mathbf{C}\mathbf{C}^{-1}\mathbf{l} = \mathbf{0}$$

$$\mathbf{l}^{\mathrm{T}}\mathbf{C}^{-\mathrm{T}}\mathbf{l} = \mathbf{0} \rightarrow \mathbf{l}^{\mathrm{T}}\mathbf{C}^{-1}\mathbf{l} = \mathbf{0} \rightarrow \mathbf{C}^{*} = \mathbf{C}^{-1}$$

(hint: **C** is symmetric)

already assume $\mathbf{l}^{\mathrm{T}}\mathbf{C}^{*}\mathbf{l} = 0$

Degenerate conics

- A conic is degenerated if matrix **C** is not a full rank matrix

e.g. two lines (rank 2)

$$\mathbf{C} = \mathbf{l}\mathbf{m}^{\mathrm{T}} + \mathbf{m}\mathbf{l}^{\mathrm{T}}$$

(**l** and **m** are line equations)

e.g. one line (rank 1)

$$\mathbf{C} = \mathbf{l}\mathbf{l}^{\mathrm{T}}$$

Example of rank 2

$\mathbf{x}^{\mathrm{T}}\mathbf{C}\mathbf{x} = \mathbf{0}$

$\mathbf{x}^{\mathrm{T}}(\mathbf{l}\mathbf{m}^{\mathrm{T}} + \mathbf{m}\mathbf{l}^{\mathrm{T}})\mathbf{x} = \mathbf{0} \rightarrow \mathbf{x}^{\mathrm{T}}(\mathbf{l}\mathbf{m}^{\mathrm{T}})\mathbf{x} + \mathbf{x}^{\mathrm{T}}(\mathbf{m}\mathbf{l}^{\mathrm{T}})\mathbf{x} = \mathbf{0}$

$(\mathbf{x}^{\mathrm{T}}\mathbf{l})(\mathbf{m}^{\mathrm{T}}\mathbf{x}) + (\mathbf{x}^{\mathrm{T}}\mathbf{m})(\mathbf{l}^{\mathrm{T}}\mathbf{x}) = \mathbf{0}$

We get: $\mathbf{x}^{\mathrm{T}}\mathbf{l}$ and $\mathbf{x}^{\mathrm{T}}\mathbf{m}$ must be zero.

Example of rank 1

$$\mathbf{l} = \begin{bmatrix}1\\2\\3\end{bmatrix}, \mathbf{C} = \begin{bmatrix}1\\2\\3\end{bmatrix} \cdot \begin{bmatrix}1 & 2 & 3\end{bmatrix} = \begin{bmatrix}1 & 2 & 3\\2 & 4 & 6\\3 & 6 & 9\end{bmatrix}$$

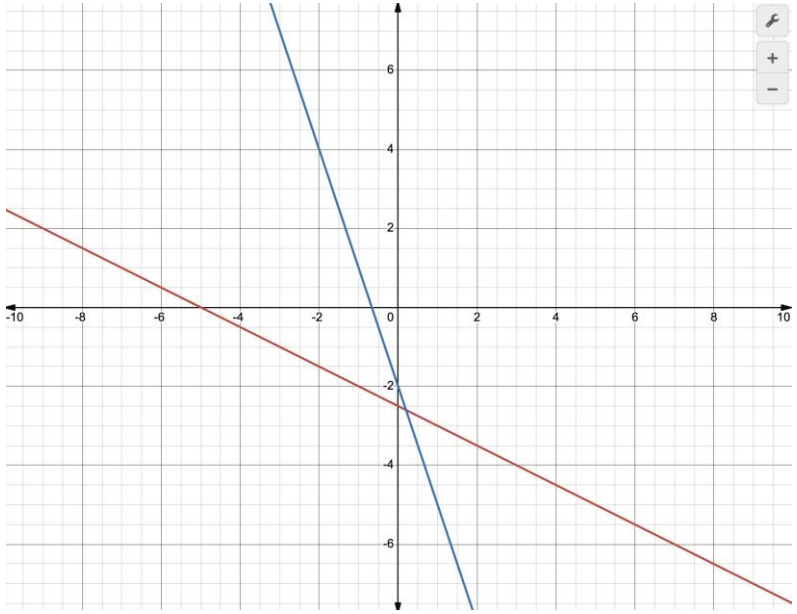
Degenerate line conics: 2 points (rank 2), double point (rank1)

Degenerate conics (example)

- Example (rank 2)

$$\mathbf{C} = \mathbf{l}\mathbf{m}^{\mathrm{T}} + \mathbf{m}\mathbf{l}^{\mathrm{T}}$$

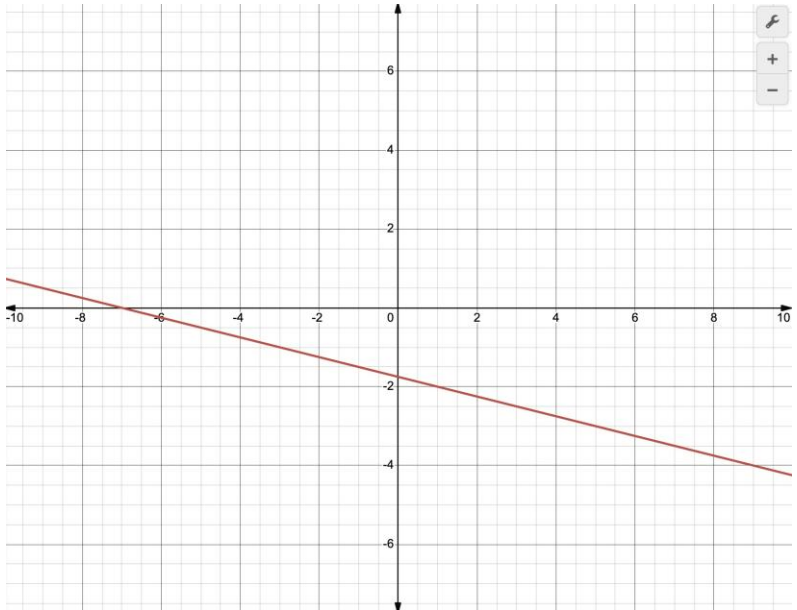
$$\mathbf{l} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}, \mathbf{l} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} \rightarrow \mathbf{C} = \begin{bmatrix} 6 & 7 & 17 \\ 7 & 4 & 9 \\ 17 & 9 & 20 \end{bmatrix}$$



- Example (rank 1)

$$\mathbf{C} = \mathbf{l}\mathbf{l}^{\mathrm{T}}$$

$$\mathbf{l} = \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix} \rightarrow \mathbf{C} = \begin{bmatrix} 1 & 4 & 7 \\ 4 & 16 & 28 \\ 7 & 28 & 49 \end{bmatrix}$$



Projective transformations

- Definition:

A *projectivity* is an invertible mapping ***h*** from P^2 to itself such that three points x_1,x_2,x_3 lie on the same line if and only if $h(x_1),h(x_2),h(x_3)$ do.

- Theorem:

A mapping $\boldsymbol{h}:P^2\rightarrow P^2$ is a projectivity if and only if there exists a non-singular 3x3 matrix **H** such that for any point in P^2 represented by a vector **x** it is true that $\boldsymbol{h}(\mathbf{x})=\mathbf{Hx}$

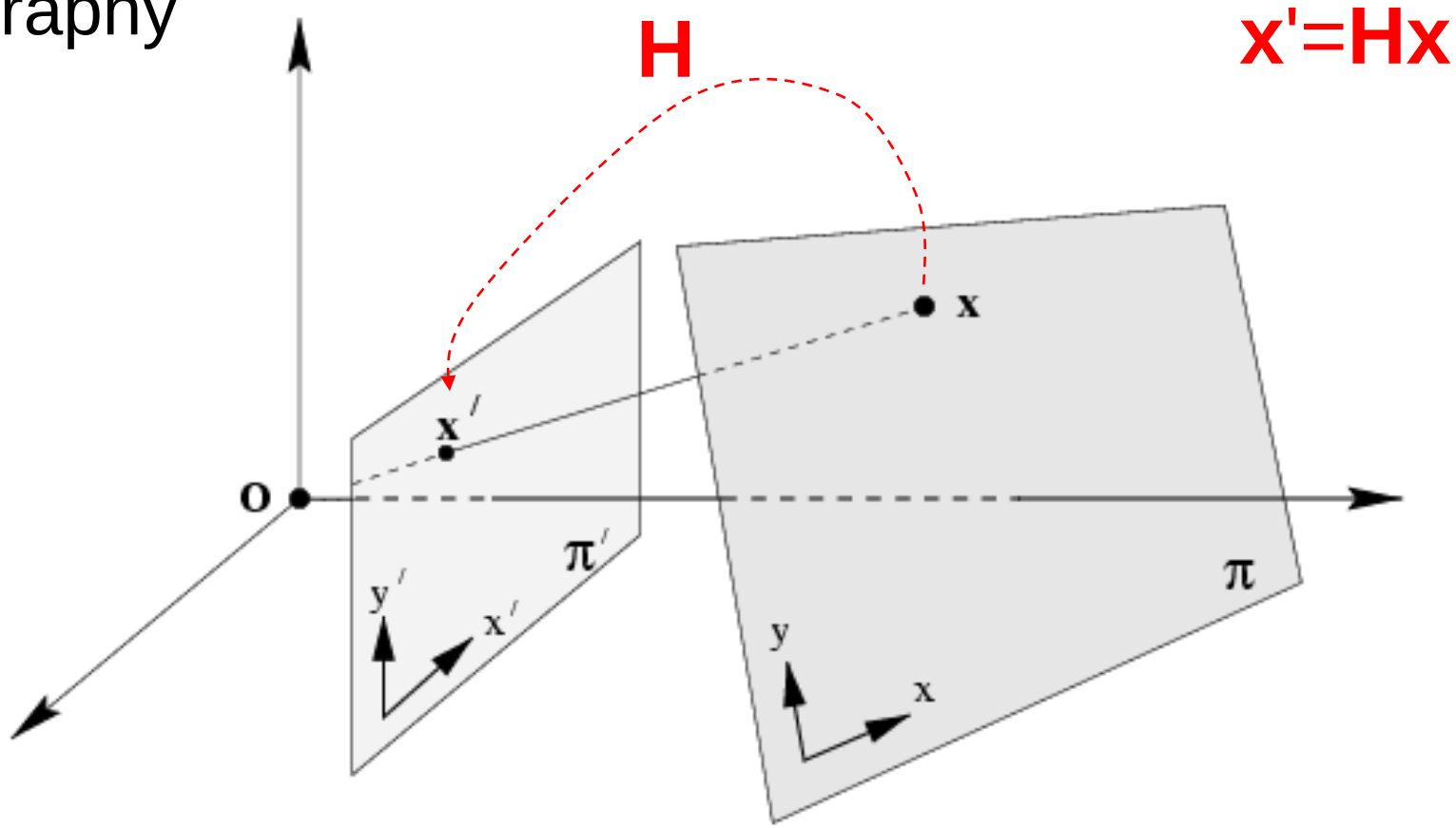
- Definition: Projective transformation

$$\begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \text{or} \quad \mathbf{x}' = \mathbf{Hx} \quad \text{(8DOF)}$$

- Projectivity = Collineation = Projective Transformation = Homography

Application: mapping between planes

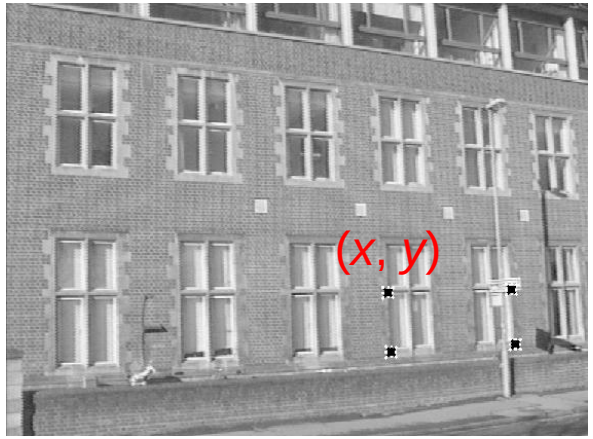
- Homography



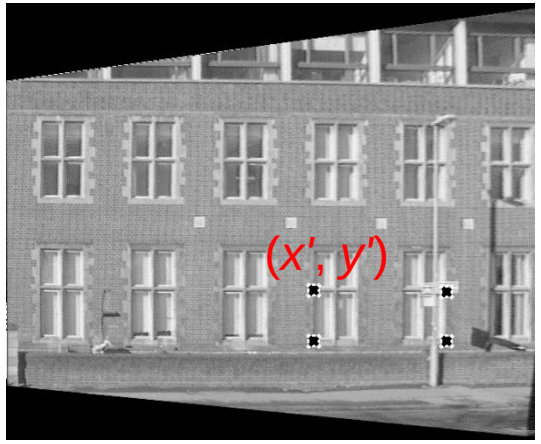
central projection may be expressed by **x'=Hx**

Homography: to remove projective effect

Input



Output



$$\begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$x' = \frac{x_1'}{x_3'} = \frac{h_{11}x+h_{12}y+h_{13}}{h_{31}x+h_{32}y+h_{33}} \quad \text{and} \quad y' = \frac{x_2'}{x_3'} = \frac{h_{21}x+h_{22}y+h_{23}}{h_{31}x+h_{32}y+h_{33}}$$

$$\rightarrow \begin{cases} x'(h_{31}x + h_{32}y + h_{33}) = h_{11}x + h_{12}y + h_{13} \\ y'(h_{31}x + h_{32}y + h_{33}) = h_{21}x + h_{22}y + h_{23} \end{cases}$$

(linear in h_{ij})

2 constraints/point, 8DOF $\Rightarrow$ 4 points needed)

Note! NO calibration at all necessary, better ways to compute

Homography: to remove projective effect

- Rewrite equation

$$\begin{cases} xh_{11} + yh_{12} + h_{13} - x'xh_{31} - x'yh_{32} - x'h_{33} = 0 \\ xh_{21} + yh_{22} + h_{23} - y'xh_{31} - y'yh_{32} - y'h_{33} = 0 \end{cases}$$

- Normalize h_{ij} with h_{33}, (replace h_{ij}/h_{33} with h_{ij} temporarily, set h_{33}=1)

$$\begin{cases} xh_{11} + yh_{12} + h_{13} - x'xh_{31} - x'yh_{32} = x' \\ xh_{21} + yh_{22} + h_{23} - y'xh_{31} - y'yh_{32} = y' \end{cases}$$

- In matrix form:

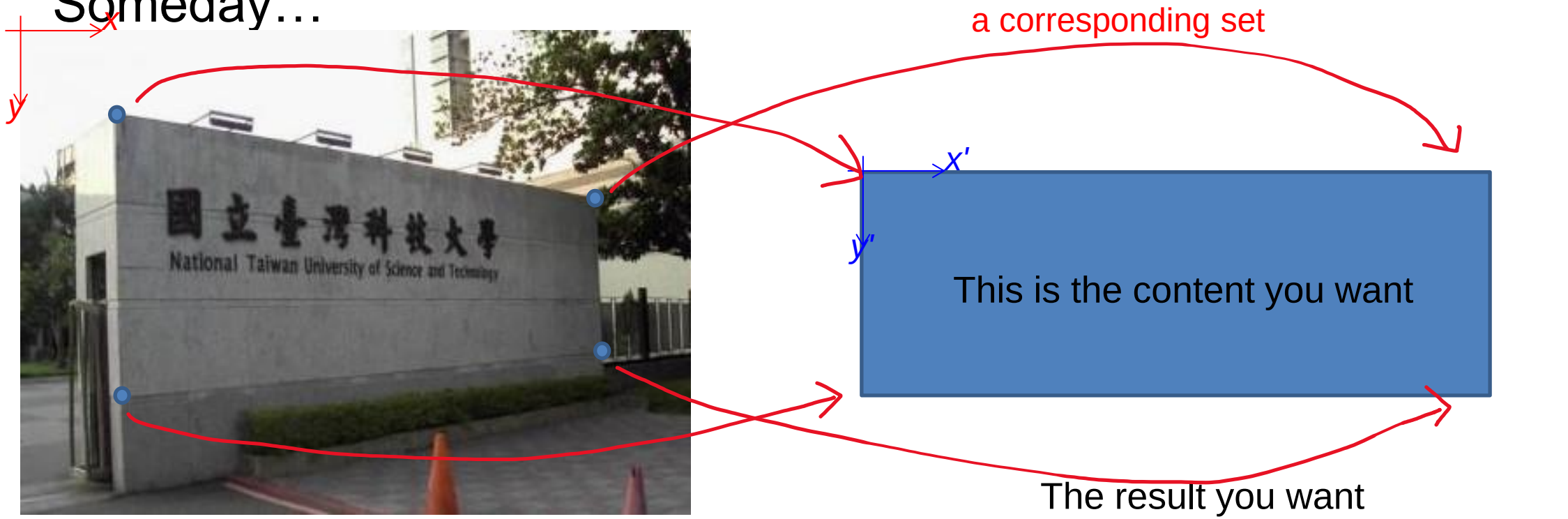
$$\begin{bmatrix} x & y & 1 & 0 & 0 & 0 & -x'x & -x'y \\ 0 & 0 & 0 & x & y & 1 & -y'x & -y'y \end{bmatrix} \begin{bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{21} \\ h_{22} \\ h_{23} \\ h_{31} \\ h_{32} \end{bmatrix} = \begin{bmatrix} x' \\ y' \end{bmatrix}$$

If you have one set of correspondence, you get (x, y) & (x', y'). So, you need at least four correspondences for solving 8 unknowns. Note: here, one correspondence forms two equations

8 unknowns to be solved

Homography: to remove projective effect

- For example: Take a picture, then remove the distortion. Someday…



A picture you took

The result you want
(This is normally defined in your application)

Define your problem, first!!!

Homography: to remove projective effect

- For example, —cont.

Correspondence:
(54,45)→(0,0)
(58,196)→(0,100)
(332,172)→(400,100)
(329,91)→(400,0)
Then, find **H**

$$\begin{bmatrix} 54 & 45 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 54 & 45 & 1 & 0 & 0 \\ 58 & 196 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 58 & 196 & 1 & -5800 & -19600 \\ 332 & 172 & 1 & 0 & 0 & 0 & -132800 & -68800 \\ 0 & 0 & 0 & 332 & 172 & 1 & -33200 & -17200 \\ 329 & 91 & 1 & 0 & 0 & 0 & -131600 & -36400 \\ 0 & 0 & 0 & 329 & 91 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{21} \\ h_{22} \\ h_{23} \\ h_{31} \\ h_{32} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 100 \\ 400 \\ 100 \\ 400 \\ 0 \end{bmatrix}$$

Homography: to remove projective effect

- For example, —cont.

$$\begin{bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{21} \\ h_{22} \\ h_{23} \\ h_{31} \\ h_{32} \end{bmatrix} = \begin{bmatrix} 54 & 45 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 54 & 45 & 1 & 0 & 0 \\ 58 & 196 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 58 & 196 & 1 & -5800 & -19600 \\ 332 & 172 & 1 & 0 & 0 & 0 & -132800 & -68800 \\ 0 & 0 & 0 & 332 & 172 & 1 & -33200 & -17200 \\ 329 & 91 & 1 & 0 & 0 & 0 & -131600 & -36400 \\ 0 & 0 & 0 & 329 & 91 & 1 & 0 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 100 \\ 400 \\ 100 \\ 400 \\ 0 \end{bmatrix}$$

$$= \begin{matrix} -0.0030 & -0.0086 & -0.0007 & 0.0085 & -0.0008 & -0.0160 & 0.0045 & 0.0160 \\ -0.0065 & 0.0002 & 0.0066 & -0.0002 & 0.0000 & 0.0004 & -0.0001 & -0.0004 \\ 1.4550 & 0.4537 & -0.2620 & -0.4511 & 0.0425 & 0.8424 & -0.2355 & -0.8450 \\ 0.0003 & -0.0024 & -0.0003 & -0.0012 & 0.0006 & 0.0002 & -0.0006 & 0.0035 \\ -0.0020 & -0.0072 & 0.0020 & 0.0073 & -0.0038 & -0.0012 & 0.0038 & 0.0011 \\ 0.0735 & 1.4550 & -0.0730 & -0.2620 & 0.1364 & 0.0425 & -0.1368 & -0.2355 \\ 0.0000 & -0.0000 & -0.0000 & 0.0000 & 0.0000 & -0.0000 & -0.0000 & 0.0000 \\ -0.0000 & 0.0000 & 0.0000 & -0.0000 & -0.0000 & 0.0000 & 0.0000 & -0.0000 \end{matrix} * \begin{matrix} 0 \\ 0 \\ 0 \\ 100 \\ 400 \\ 100 \\ 400 \\ 0 \end{matrix} = \begin{matrix} 0.7210 \\ -0.0191 \\ -38.0771 \\ -0.1029 \\ 0.6150 \\ -22.1199 \\ -0.0016 \\ 0.0001 \end{matrix}$$

$$\mathbf{H} = \begin{bmatrix} 0.721 & -0.0191 & -38.0771 \\ -0.1029 & 0.6150 & -22.1199 \\ -0.0016 & 0.0001 & 1 \end{bmatrix}$$

Homography: to remove projective effect

- For example, —cont.

	p1 =	p2 =	p3 =	p4 =
original points:	54 45 1	58 196 1	332 172 1	329 91 1
	>> H*p1	>> H*p2	>> H*p3	>> H*p4
	ans =	ans =	ans =	ans =
x'=Hx	0 0.0000 0.9192	-0.0000 92.4536 0.9245	198.0257 49.5064 0.4951	197.4097 0.0000 0.4935
normalized	0 0.0000 1.0000	-0.0000 100.0000 1.0000	400.0000 100.0000 1.0000	400.0000 0.0000 1.0000
Desired points:	$(0,0,1)^T$	$(0,100,1)^T$	$(400,100,1)^T$	$(400, 0,1)^T$

Homography: to remove projective effect

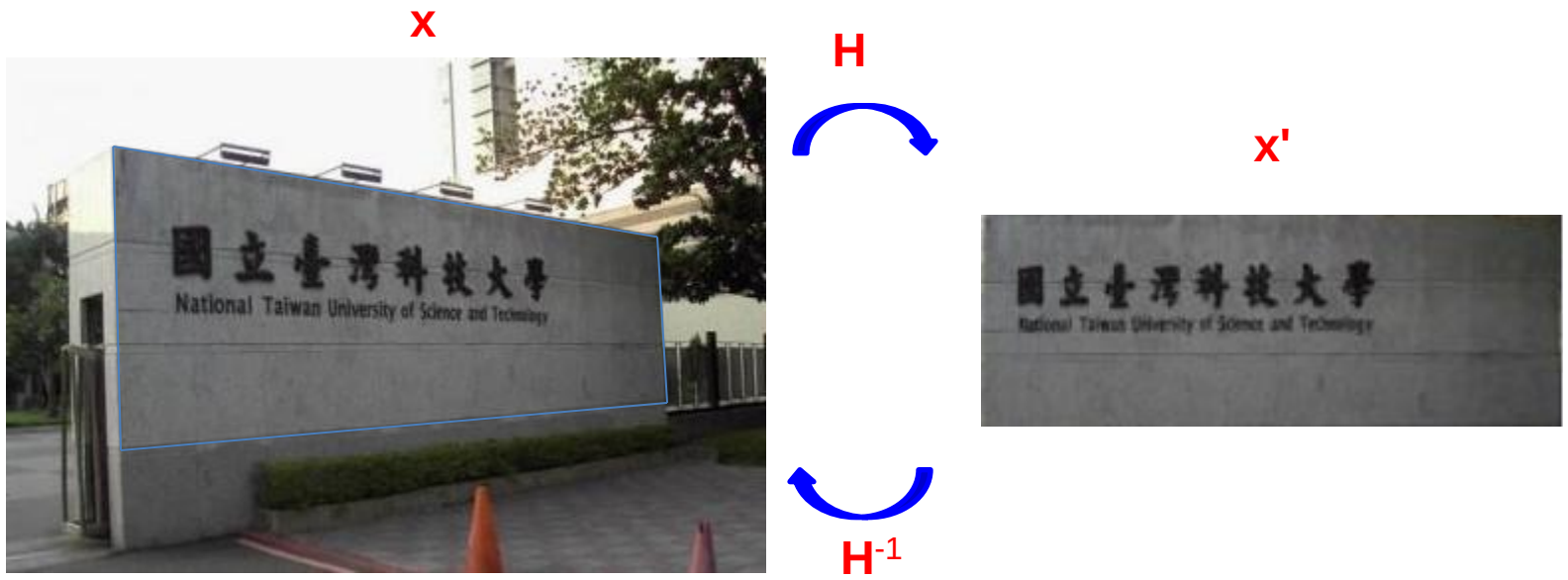
- For example, —cont. (inverse mapping)

	pp1 =	pp2 =	pp3 =	pp4 =
desired points:	0 0 1	0 100 1	400 100 1	400 0 1
$x=H^{-1}x'$	>> inv(H)*pp1 ans = 58.7480 48.9567 1.0879	>> inv(H)*pp2 ans = 62.7342 211.9982 1.0816	>> inv(H)*pp3 ans = 670.6200 347.4296 2.0199	>> inv(H)*pp4 ans = 666.6338 184.3881 2.0262
normalized	54.0000 45.0000 1.0000	58.0000 196.0000 1.0000	332.0000 172.0000 1.0000	329.0000 91.0000 1.0000

Original points: $(54,45,1)^T$ $(58,196,1)^T$ $(332,172,1)^T$ $(329,91,1)^T$

Homography: to remove projective effect

- For example, —cont. (inverse mapping)



Filling correct COLOR:
You are knowing to filling color in a “400x100” image. For each pixel, you need to calculate its color by applying H^{-1} to its coordinate.

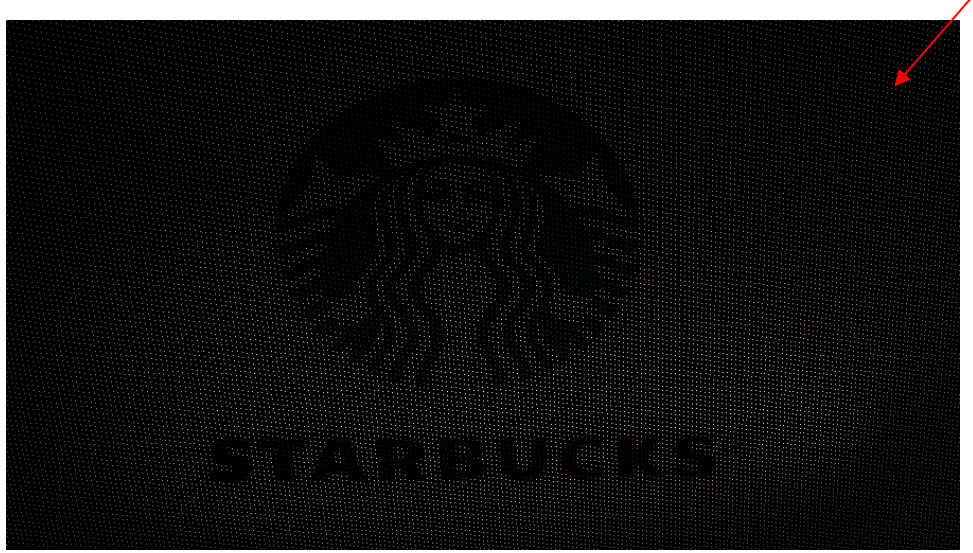
Homography: filling pixels

320x240

H

900x500

Missing pixels



```
#mapping from left to right
for i in range(320)
    for j in range(240)
        BigImg = H*SmallImg
```

Homography: filling pixels

320x240

H'

900x500

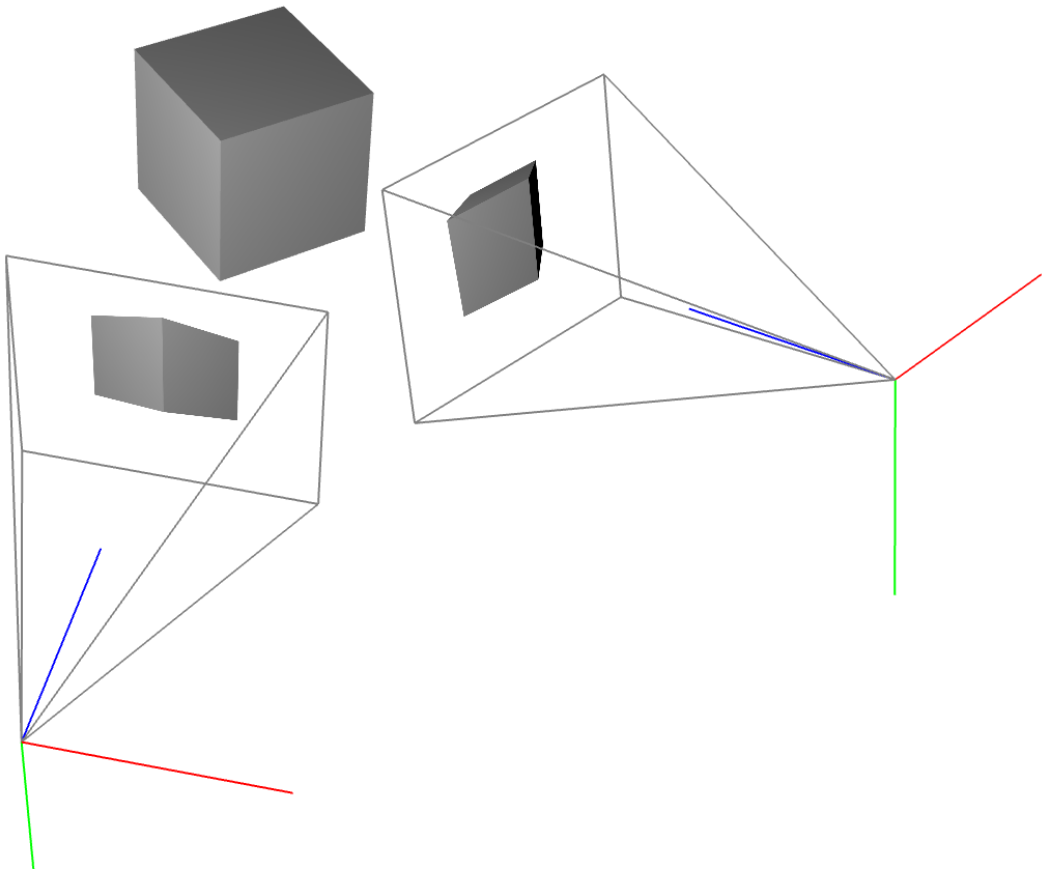
```
#query from left
for i in range(900)
    for j in range(500)
        pixelVal = H'*BigImg
        BigImg[][] = pixelVal
```

Be better if can do additional interpolation

Homography: What is it?

If you have at least 4 corresponding points, a homography can dominate the transformation between two images. So, you do NOT need to determine **K**[**R**|**t**] for 2 views

(Note: ONLY planar structure in 3D)



Homography in OpenCV

- openCV provides various kinds of mapping operations in computer vision.

Correspondence:
(54,45)→(0,0)
(58,196)→(0,100)
(332,172)→(400,100)
(329,91)→(400,0)
Then, find **H**

Sample Code (in openCV):
findHomography

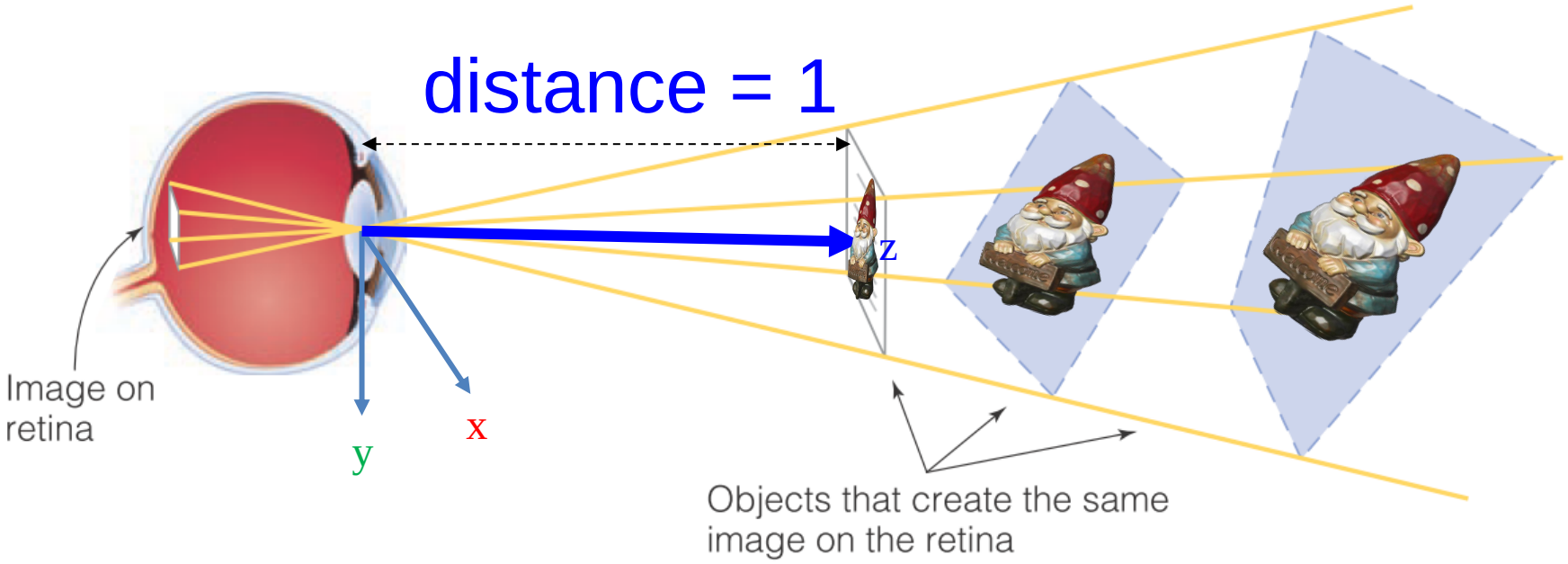
Source Points:
54.000000 45.000000
58.000000 196.000000
332.000000 172.000000
329.000000 91.000000

Destination Points:
0.000000 0.000000
0.000000 100.000000
400.000000 100.000000
400.000000 0.000000

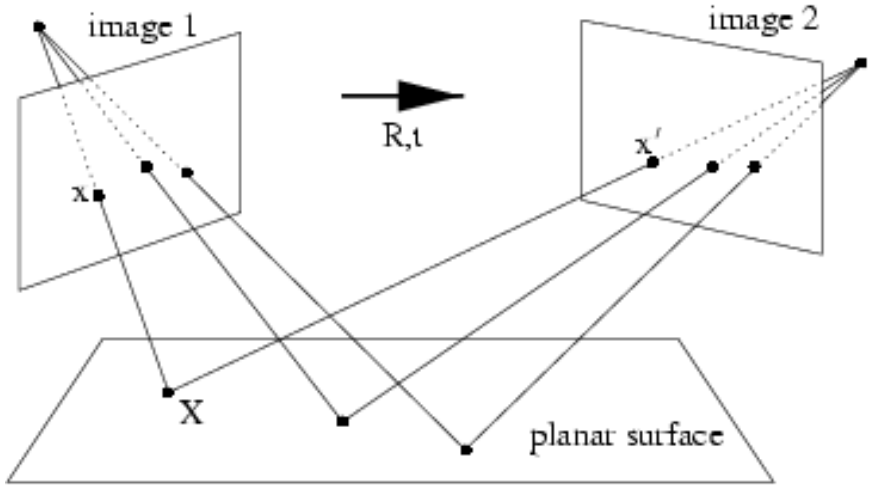
Homography Matrix:
0.721049 -0.019101 -38.077095
-0.102873 0.615001 -22.119894
-0.001561 0.000077 1.000000

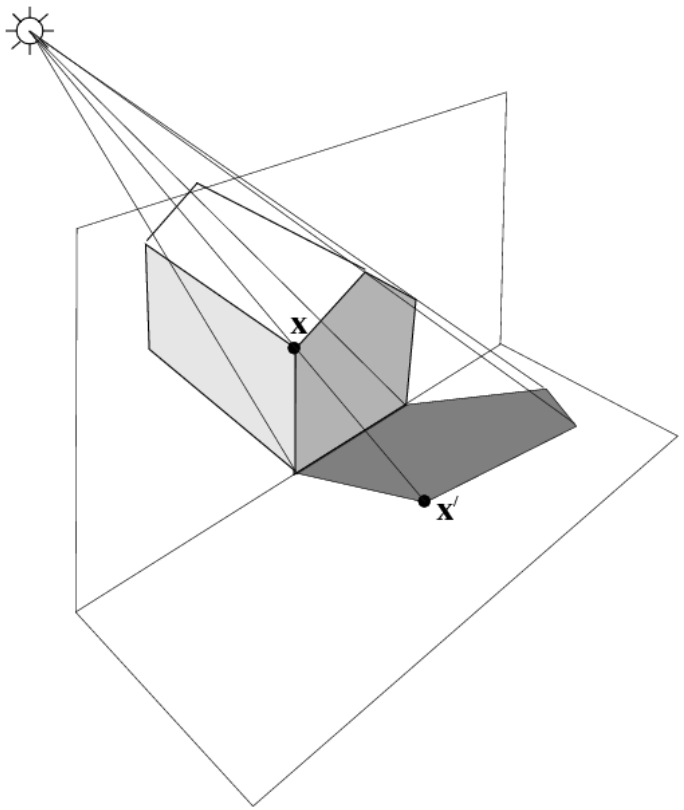
https://docs.opencv.org/3.4/d9/d0c/group__calib3d.html#ga4abc2ece9fab9398f2e560d53c8c9780

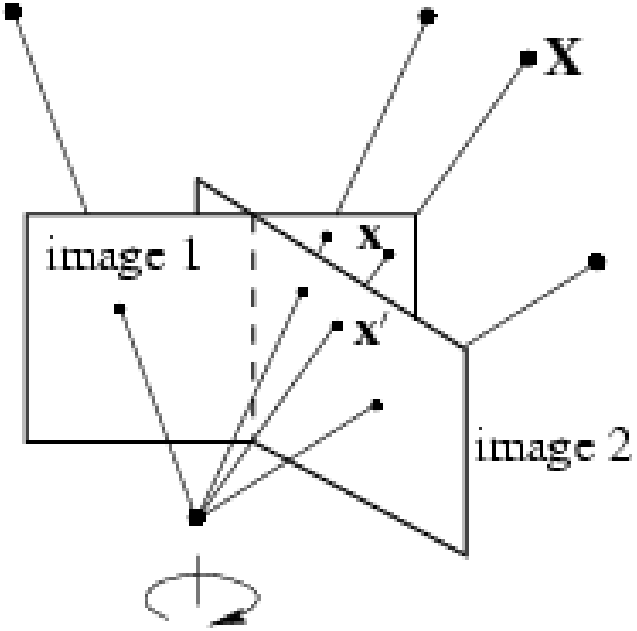
Homography: a projection conversion



Homography: more examples







Transformation of lines and conics

Homography for specific shape

- For a point transformation

$$\mathbf{x}' = \mathbf{H}\mathbf{x}$$

- Transformation for lines

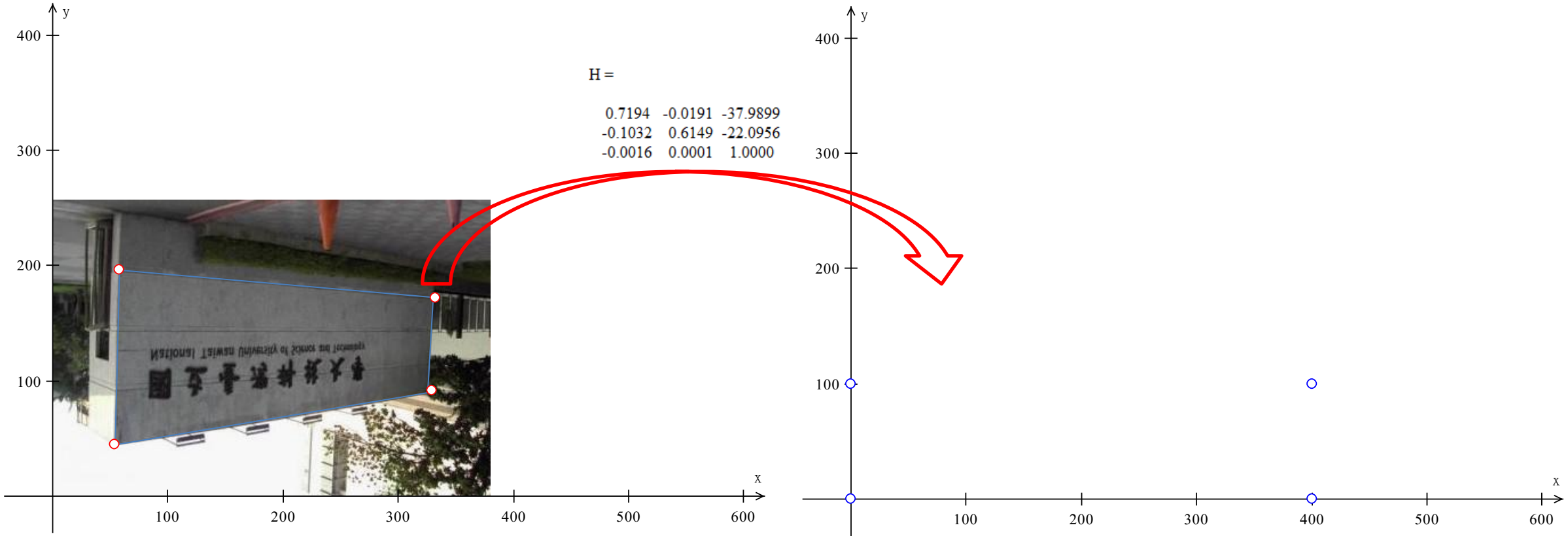
$$\mathbf{l}' = \mathbf{H}^{-\mathrm{T}}\mathbf{l}$$

- Transformation for conics

$$\mathbf{C}' = \mathbf{H}^{-\mathrm{T}}\mathbf{C}\mathbf{H}^{-1}$$

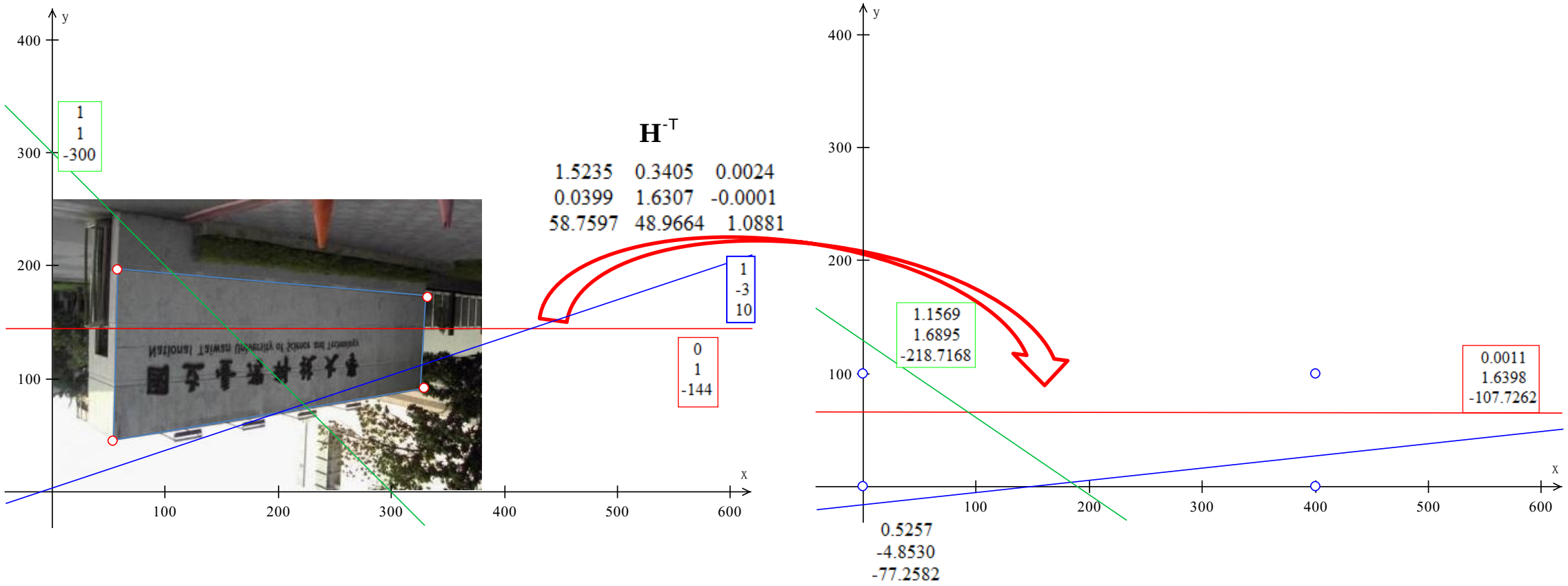
Transformation of lines and conics

- Example: (the same with previous, but mirror for convenience)



Transformation of lines and conics

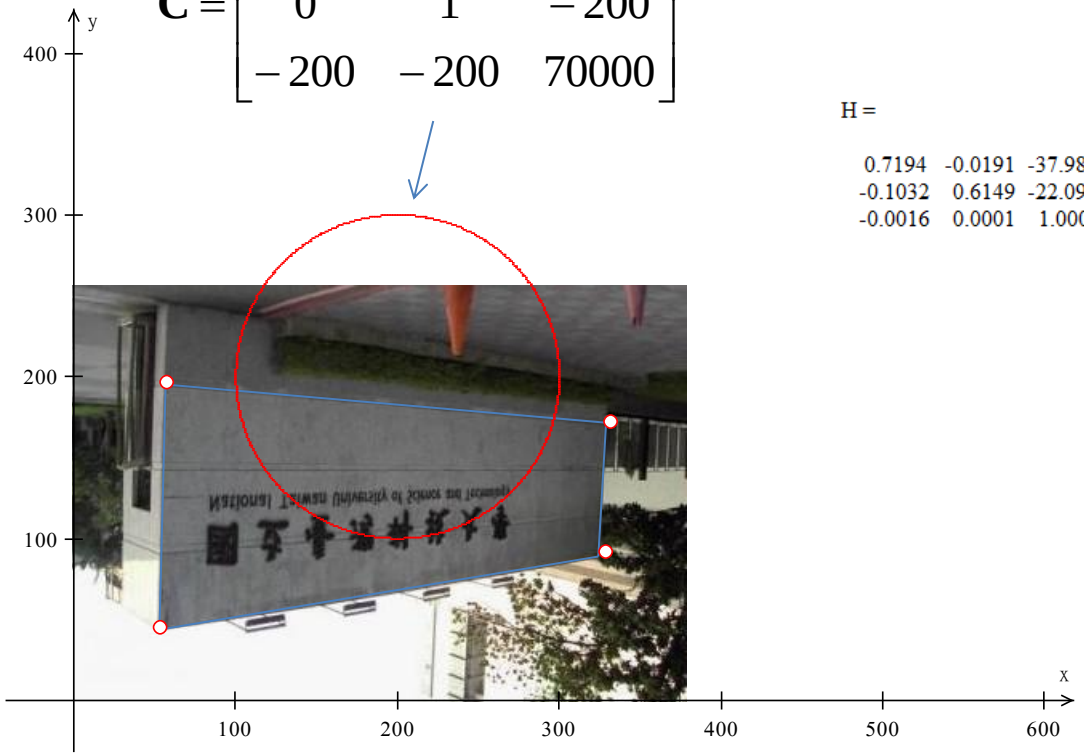
- Example: lines transformation ($\mathbf{l}' = \mathbf{H}^{-\mathrm{T}}\mathbf{l}$)



Transformation of lines and conics

- Example: conics transformation ($\mathbf{C}' = \mathbf{H}^{-\mathrm{T}}\mathbf{C}\mathbf{H}^{-1}$)

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & -200 \\ 0 & 1 & -200 \\ -200 & -200 & 70000 \end{bmatrix}$$

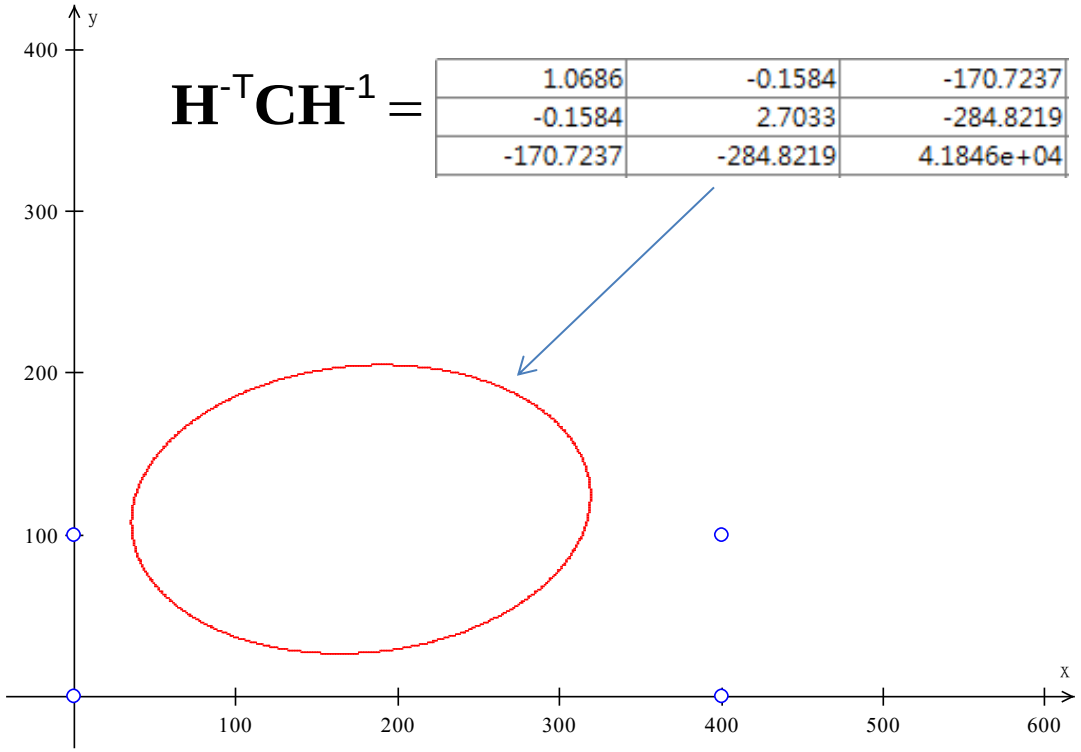


H =

0.7194	-0.0191	-37.9899
-0.1032	0.6149	-22.0956
-0.0016	0.0001	1.0000

$\mathbf{H}^{-\mathrm{T}}\mathbf{C}\mathbf{H}^{-1} =$

1.0686	-0.1584	-170.7237
-0.1584	2.7033	-284.8219
-170.7237	-284.8219	4.1846e+04



A hierarchy of transformations

- Projective linear group
  - Affine group (last row (0,0,1))
    - Euclidean group (upper left 2x2 orthogonal)
      - Oriented Euclidean group (upper left 2x2 det 1)

Alternative, characterize transformation in terms of elements or quantities that are preserved or *invariant*

e.g. Euclidean transformations leave distances unchanged

Four classic types of transformation

- Isometrics
- Similarities
- Affine mapping
- Projective mapping

Four classic types of transformation—cont.

- Class I: Isometries (*iso*=same, *metric*=measure)

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \varepsilon \cdot \cos\theta & -\sin\theta & t_x \\ \varepsilon \cdot \sin\theta & \cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}, \text{ where } \varepsilon = \pm 1$$

Same orientation: where $\varepsilon = 1$

Reversed orientation: where $\varepsilon = -1$

$$\mathbf{x}' = \mathbf{H}_E \mathbf{x} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ 0^{\mathrm{T}} & 1 \end{bmatrix} \mathbf{x}, \qquad \text{note: } \mathbf{R}^{\mathrm{T}}\mathbf{R} = \mathbf{I}$$

3DOF (1 rotation, 2 translation)

special cases: pure rotation, pure translation

Invariants: length, angle, area

Four classic types of transformation—cont.

- Class II: Similarities (*isometry* + *scale*)

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s\cdot\cos\theta & -s\cdot\sin\theta & t_x \\ s\cdot\sin\theta & s\cdot\cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$, where s is a non-zero scalar

$$\mathbf{x}' = \mathbf{H}_S\mathbf{x} = \begin{bmatrix} s\mathbf{R} & \mathbf{t} \\ 0^{\mathrm{T}} & 1 \end{bmatrix}\mathbf{x},$$ note: $\mathbf{R}^{\mathrm{T}}\mathbf{R} = \mathbf{I}$

4DOF (1 scale, 1 rotation, 2 translation)

also know as *equi-form* (shape preserving)

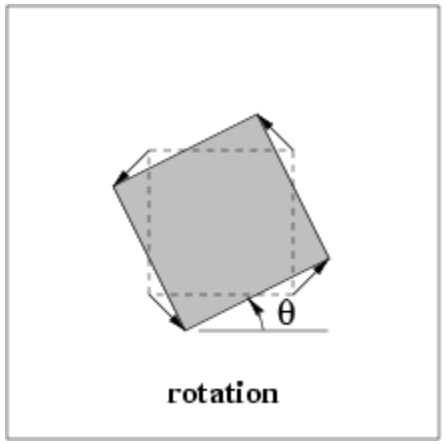
metric structure = structure up to similarity (in literature)

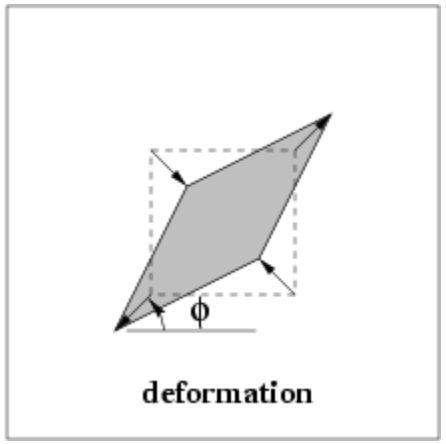
Invariants: ratios of length, angle, ratios of areas, parallel lines

Four classic types of transformation—cont.

- Class III: Affine transformations

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$





$$\mathbf{x}' = \mathbf{H}_A \mathbf{x} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ 0^{\mathrm{T}} & 1 \end{bmatrix} \mathbf{x},$$ note: $\mathbf{R}^{\mathrm{T}}\mathbf{R} = \mathbf{I}$

$$\mathbf{A} = \mathbf{R}(\boldsymbol{\theta})\mathbf{R}(\boldsymbol{\phi})\mathbf{D}\mathbf{R}(\boldsymbol{\phi}),$$ where $\mathbf{D} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$

6DOF (2 scale, 2 rotation, 2 translation)

non-isotropic scaling! (2DOF: scale ratio and orientation)

Invariants: parallel lines, ratios of parallel lengths, ratios of areas

Four classic types of transformation—cont.

- Class IV: Projective transformations

$$\mathbf{x}' = \mathbf{H}_P\mathbf{x} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{v}^{\mathrm{T}} & v \end{bmatrix}\mathbf{x}, \quad \text{where } \mathbf{v}^{\mathrm{T}} = (v_1, v_2)$$

8DOF (2 scale, 2 rotation, 2 translation, 2 line at infinity)

Action non-homogeneous over the plane

Invariants: cross-ratio of four points on a line (ratio of ratio)

Four classic types of transformation—cont.

- Action of affinities and projectivities on line at infinity

$$\begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0}^{\mathrm{T}} & v \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix} = \begin{bmatrix} \mathbf{A} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ 0 \end{bmatrix}$$

Line at infinity stays at infinity,
but points move along line

$$\begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{v}^{\mathrm{T}} & v \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix} = \begin{bmatrix} \mathbf{A} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ v_1 x_1 + v_2 x_2 \end{bmatrix}$$

Line at infinity becomes finite,
allows to observe vanishing points, horizon,

Four classic types of transformation—cont.

- Decomposition of projective transformations

$$\mathbf{H} = \mathbf{H}_S \mathbf{H}_A \mathbf{H}_P = \begin{bmatrix} s\mathbf{R} & \mathbf{t} \\ \mathbf{0}^{\mathrm{T}} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{K} & \mathbf{0} \\ \mathbf{0}^{\mathrm{T}} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{v}^{\mathrm{T}} & v \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{v}^{\mathrm{T}} & v \end{bmatrix}$$

Similarity

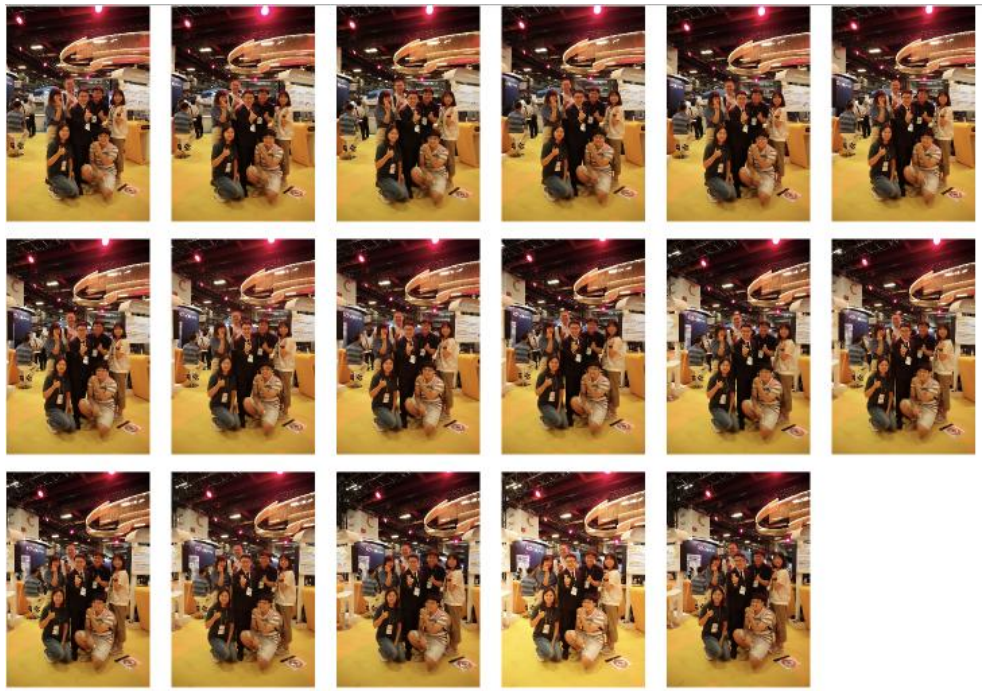
Affine

Projective

Application in homography decomposition

Multiple view images

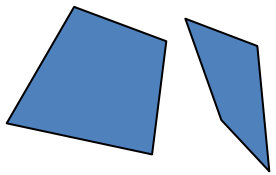
- Perform quality alignment
- Preseve property of “perspective effect”



Overview transformations

Projective
8 DOF

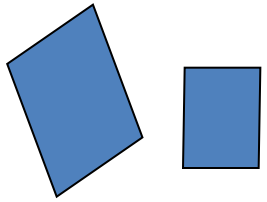
$$\begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$$



Concurrency, collinearity, order of contact (intersection, tangency, inflection, etc.), cross ratio

Affine
6 DOF

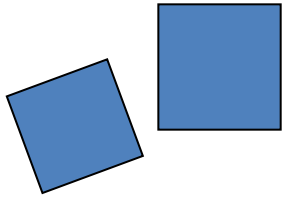
$$\begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$$



Parallellism, ratio of areas, ratio of lengths on parallel lines (e.g midpoints), linear combinations of vectors (centroids).
The line at infinity l_∞

Similarity
4 DOF

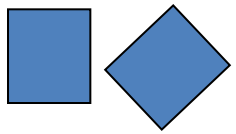
$$\begin{bmatrix} sr_{11} & sr_{12} & t_x \\ sr_{21} & sr_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$$



Ratios of lengths, angles.
The circular points I,J

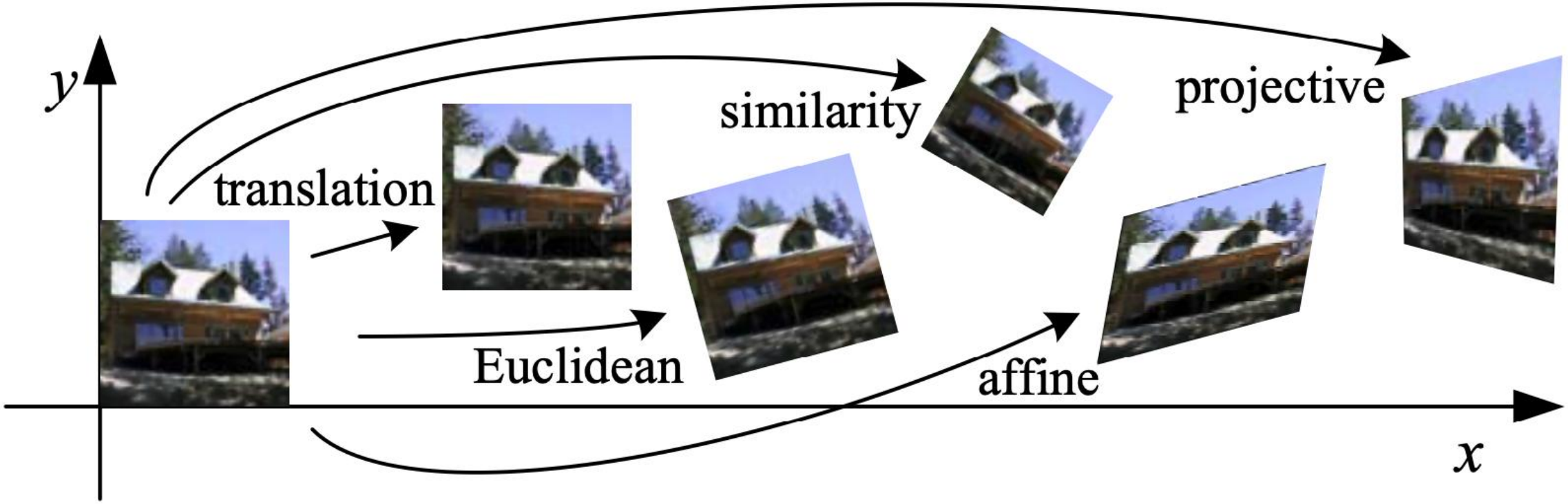
Euclidean
3 DOF

$$\begin{bmatrix} r_{11} & r_{12} & t_x \\ r_{21} & r_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$$



lengths, areas.

Overview transformations



L. Nucci, D. Narvaez, and T. Krettenauer, *Computer Vision: Algorithms and Applications*, in Dairy Science & Technology, CRC Taylor & Francis Group(June) (2022).

Number of invariants?

- The number of functional invariants is equal to, or greater than, the number of degrees of freedom of the configuration less the number of degrees of freedom of the transformation
- e.g. configuration of 4 points in general position has 8 DOF (2/pt) and so 4 similarity, 2 affinity and zero projective invariants

Summary

- Points and lines

$$\mathbf{x}^{\mathrm{T}}\mathbf{l} = \mathbf{0}\ , \quad \mathbf{l}^{\mathrm{T}}\mathbf{x} = \mathbf{0}\ , \mathbf{x} = \mathbf{l} \times \mathbf{l}^{'}, \mathbf{l} = \mathbf{x} \times \mathbf{x}^{'}$$

$$\mathbf{l}_{\infty} = (0, 0, 1)^{\mathrm{T}}, \qquad \mathbf{x}_{\infty} = (x, y, 0)^{\mathrm{T}}$$

- Conics and dual conics

$$\mathbf{x}^{\mathrm{T}}\mathbf{C}\mathbf{x} = 0, \quad \mathbf{l}^{\mathrm{T}}\mathbf{C}^{*}\mathbf{l} = 0, \quad \mathbf{C}^{*} = \mathbf{C}^{-1}, \quad \mathbf{l} = \mathbf{C}\mathbf{x}$$

- Projective transformations

$$\mathbf{x}^{'} = \mathbf{H}\mathbf{x}, \quad \mathbf{l}' = \mathbf{H}^{-\mathrm{T}}\mathbf{l}$$

$$\mathbf{C}^{'} = \mathbf{H}^{-\mathrm{T}}\mathbf{C}\mathbf{H}^{-1}$$

Q & A

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